

Cryptocurrency Volume-Weighted Time Series Momentum

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Abstract

This paper finds strong and significant evidence of cryptocurrency TSMOM using volume-weighted market return. Based on the TSMOM strategy, we show that volume-weighted winner minus loser portfolios can generate significant returns which outperform other TSMOM strategies. Specifically, our strategy can gain 0.94% per day with an annualized Sharpe ratio of 2.17. Our findings cannot be fully explained by market, size, and momentum factors and are not subsumed by cross-sectional and MAX momentum returns, and is not the result of wash trading volume. Our findings suggest that trading volume may provide signals within cryptocurrency markets to create profitable strategies.

Keywords: Cryptocurrency; Time series momentum; Trading volume; Return predictability

JEL Classification: G11; G12

1. Introduction

Time series momentum (Moskowitz et al., 2012), the effect of return continuation of individual assets, has been shown to exist across many asset classes, such as stocks (He and Li, 2015, Lim et al., 2018, Hutchinson and O’Brien, 2020, Huang et al., 2020), currencies (Zhang, 2022), and across assets (Pitkäjärvi et al., 2020, Fernandez-Perez et al., 2023). However, the related literature on a new investment vehicle, cryptocurrencies, remains limited. There is no consistent conclusion on the effect of time series momentum (TSMOM) in the cryptocurrency market in recent studies, with varied results among different studies. With the growing market size and trading volume of cryptocurrencies, we re-examine the TSMOM effect and provide evidence of the profitability of the TSMOM strategy in relation to trading volume. Specifically, we document robust and notable TSMOM in volume-weighted cryptocurrency market return and construct portfolios by volume-weighted TSMOM strategy (VW-WML). We show that the strategy return outperforms the traditional TSMOM strategy and cannot be fully explained by the cryptocurrency three-factor model proposed by Liu et al. (2022).

Our primary contributions are twofold. First, we add evidence to the literature on TSMOM in the cryptocurrency market (Borgards, 2021, Grobys and Sapkota, 2019, Kosciuszko et al., 2019, Liu and Tsyvinski, 2021). We find a strong TSMOM effect in the cryptocurrency market by analyzing the volume-weighted market return. Contrary to Liu and Tsyvinski (2021), we do not use capital-weighted market return because the market capitalization does not represent genuine trading activities in the cryptocurrency market since a coin’s market capitalization can be manipulated by founders minting new coins whenever they wish.¹ For example, in our research sample, some coins have a market capitalization of more than \$1 million, but a daily trading volume of no

¹<https://medium.com/blockchain-review/8deb78c50995>

more than \$10,000 per day. Hence, the trading volume can better represent a cryptocurrency’s popularity and trading activities. Furthermore, the important linkage between trading volume and momentum payoffs is uncovered in Lee and Swaminathan (2000). The authors document that lower past volume, proxied by the stock’s turnover rate, tends to generate higher future returns. Lee and Swaminathan (2000) focus on cross-sectional momentum and the stock market. We aim to provide evidence on the volume-weighted TSMOM strategy and show a strong and significant TSMOM effect. Nevertheless, Cong et al. (2023) and Aloosh and Li (2024) document the suspicion of cryptocurrency wash trading, indicating the potential fake transactions conducted by exchanges. To mitigate concerns related to wash trading, we conduct robustness checks with volume reported from trusted exchanges. The results yield qualitatively very consistent results, affirming the presence of a significant TSMOM effect and the profitability of the strategy.

Second, we contribute to the studies of the TSMOM trading strategy in the cryptocurrency market (Gerritsen et al., 2020, Tzouvanas et al., 2020) by adjusting the TSMOM strategy with trading volume. We construct a volume-weighted Winner Minus Loser (VW-WML) portfolio. The winner portfolio includes coins with positive past days’ return, and the loser portfolio comprises coins with negative formation days’ return. Within winner and loser portfolios, each coin is weighted by daily trading volume. Subsequently, the VW-WML portfolio is constructed by taking a long position on the winner portfolio and a short position on the loser portfolio. Following Lim et al. (2018), we compare the VW-WML portfolio return to other weighting schemes, including capital-weighted, equally-weighted, and inverse volatility-weighted. We show that the VW-WML portfolio generates significantly positive returns and outperforms other portfolios and the market return, demonstrating that the TSMOM strategy weighted by trading volume is more profitable among common adjusting options, such as market capitalization, equal proportion, and inverse volatility. Moreover, we find that the VW-WML return cannot be fully explained by the well-

known cryptocurrency three-factor model introduced by Liu et al. (2022). Additionally, we show that the VW-WML portfolio profit is not subsumed by cross-sectional momentum (Jegadeesh and Titman, 1993) or MAX momentum (Li et al., 2021).

Our study extends the literature on time series momentum, which was first introduced by Moskowitz et al. (2012) documenting the return persistence of individual assets. TSMOM strategy consists of purchasing assets with positive past returns while selling assets with negative past returns. The strategy focuses on the signal of an individual asset instead of the relative strength between assets, differentiating it from the cross-sectional momentum. Previous literature focuses on the existence of TSMOM of asset classes (Asness et al., 2013, Huang et al., 2020, Lim et al., 2018, Moskowitz et al., 2012), the profitability of TSMOM strategy (He and Li, 2015), the intra-day TSMOM (Li et al., 2020, Shen et al., 2022), and the difference between time series momentum and cross-sectional momentum (Goyal and Jegadeesh, 2018).

There are a few studies on the time series momentum in the cryptocurrency market. Liu and Tsyvinski (2021), Liu et al. (2022), Sakkas and Urquhart (2024) find a strong TSMOM effect of cryptocurrency market return. Borgards (2021) shows strong time series momentum and applies a dynamic TSMOM strategy to generate profits while Tzouvanas et al. (2020) examine the profitability of momentum trading and find its diversification benefits. On the contrary, Grobys and Sapkota (2019) indicate no evidence of significant momentum profits. Kosciuszko et al. (2019) compare the momentum and contrarian effects among cryptocurrencies and find that the short-term contrarian effect dominates the momentum effect. Based on the previous studies, there is no consistent conclusion for the presence of time series momentum.

Furthermore, the research on the direct connection between time series momentum and cryptocurrency trading volume remains limited. In the stock market, Lee and Swaminathan (2000) link cross-sectional momentum to trading volume. They examine the volume-based price momentum

strategy, which first sorts the stocks into ten groups based on past relative performance and then sorts the stocks within each group based on the average daily volume. They show that past trading volume negatively predicts future stock returns. In the cryptocurrency market, literature stresses the relationship between trading volume and returns. Bianchi et al. (2022) reveal that the interaction between lagged returns and trading volume holds predictive information in the dynamics of cryptocurrency returns. Thus, to address this gap, we study the connection between trading volume and return continuation by analyzing the TSMOM strategy adjusted by trading volume. Recently, Fieberg et al. (2024) show a strong trend factor in cryptocurrency returns which cannot be subsumed by known factors.

In this paper, we first examine the TSMOM effect in the cryptocurrency market. By regressing the current market return on the past cumulative market return, we find a strong and significant TSMOM from 3-day to 2-week time horizons. In addition, following Liu and Tsyvinski (2021), we study the magnitude of the TSMOM effect by grouping the past cumulative market returns into terciles, including high, middle, and low, and evaluating each group’s subsequent day’s return. The differences between High and Low average returns are all positive among time horizons, indicating that higher past returns can generate higher future returns. We also apply tercile cutoffs calculated from the first 2 years of data in our sample, from 2014 to 2015, and examine the out-of-sample TSMOM magnitude, from 2016 to 2023, finding similar results.

Second, we evaluate the performance of the volume-weighted Winner Minus Loser portfolio. The VW-WML portfolio is created based on the TSMOM strategy adjusted by volume weighting. We first categorise coins into winner and loser portfolios by past returns, weighting each portfolio by trading volume. Then, we buy the winner portfolio and sell the loser portfolio. We find that the VW-WML portfolio can generate significantly positive returns, with daily average returns of 0.94% (1-day formation period), 0.87% (7-day formation period), and 0.67% (14-day

formation period). The annualised Sharpe ratios are 2.17 (1-day formation period), 2.66 (7-day formation period), and 1.86 (14-day formation period), which is high considering the volatile characteristics of the cryptocurrency market. We compare the portfolio profits to the WML portfolio with various weighting schemes as documented in Lim et al. (2018), including capital-weighting, equally-weighting, and inverse volatility-weighting. The VW-WML portfolio profits are more than twice the capital-weighted WML portfolio profits, while the equally-weighted and inverse volatility-weighted portfolio shows significant negative returns.

Third, we show that by controlling for cryptocurrency risk factors (Liu and Tsyvinski, 2021, Liu et al., 2022, Sakkas and Urquhart, 2024), including market (CMKT), size (CSMB), and momentum (CMOM) factors, the VW-WML portfolio profits cannot be well explained by the factors considered. The magnitude of the estimated alphas ranges from 0.6% to 2.2% with 1% level of significance. We further investigate the magnitude of alphas by considering transaction costs. The alphas remain significantly positive, with alphas of 0.4% to 2%. Lastly, to deal with the cryptocurrency wash trading suspicion (Aloosh and Li, 2024, Cong et al., 2023) we conduct robustness checks using volume reported by trusted exchanges from Coin Metrics data. The results are qualitatively similar to our main results, confirming that the cryptocurrency market exhibits a significant TSMOM effect.

Fourth, we show that the profitability of VW-WML is not subsumed by the cross-sectional momentum (Jegadeesh and Titman, 1993) and MAX momentum (Li et al., 2021) portfolios. Different from the TSMOM strategy, the winner and loser portfolios in the two strategies are categorised based on past relative returns (cross-sectional momentum) and the highest return during the formation periods (MAX momentum). To examine whether the profits of the VW-WML portfolio are inherited from cross-sectional and MAX momentum strategy, we regress the VW-WML return on cross-sectional and MAX momentum portfolio returns weighted by market capitalization and

trading volume. The estimated alphas are significantly positive, indicating that other momentum strategies cannot entirely explain the VW-WML portfolio profits.

Our study can benefit investors in the cryptocurrency market from several perspectives. First, the cryptocurrency market is distinct from other traditional financial assets: if we consider an equal weighting scheme, the larger coins, such as Bitcoin and Ethereum, would be under-considered. For a capital-weighting scheme, the portfolio heavily weighs on a small group of "big" coins. Market capitalization is not a suitable indicator of trading activities because of the possible manipulation of market capitalization, for instance, controlling coin scarcity by changing coin supply.² If we apply the inverse volatility-weighting scheme, the highly volatile coins would be excluded from the portfolios. It is not realistic because the investors in the market are comparably risk-lovers. We show that the TSMOM strategy works well by adjusting by volume and outperforms the normal TSMOM strategy. The evidence on the VW-WML portfolio can help traders make trading decisions and earn more profits because our portfolio is shown to outperform the market. Furthermore, the volatile and accessible cryptocurrency market also attracts speculators and retail investors, who may not hold coins for a long time compared to traditional financial assets. Our analysis based on daily trading volume can help traders better understand the market. Second, our study demonstrates that the momentum factors are predominant risks in the VW-WML portfolio compared to market and size factors. As the participation of institutional investors in the cryptocurrency market grows, our findings can provide guidance on risk management and portfolio rebalancing for them. Third, our study can aid regulatory authorities in reassessing the stability of the cryptocurrency market since the trading volume and associated returns could lead to increased speculation, potentially inflating coin prices and creating bubbles.

²<https://coinswitch.co/switch/crypto/how-does-market-cap-affect-crypto-prices/>

This paper is structured as follows: Section 2 describes the data used in this paper. Section 3 documents the TSMOM effect in volume-weighted cryptocurrency market return followed by the discussion of the TSMOM strategy profitability in Section 4. The robustness checks are reported in Section 5. Finally, the conclusion is in Section 6.

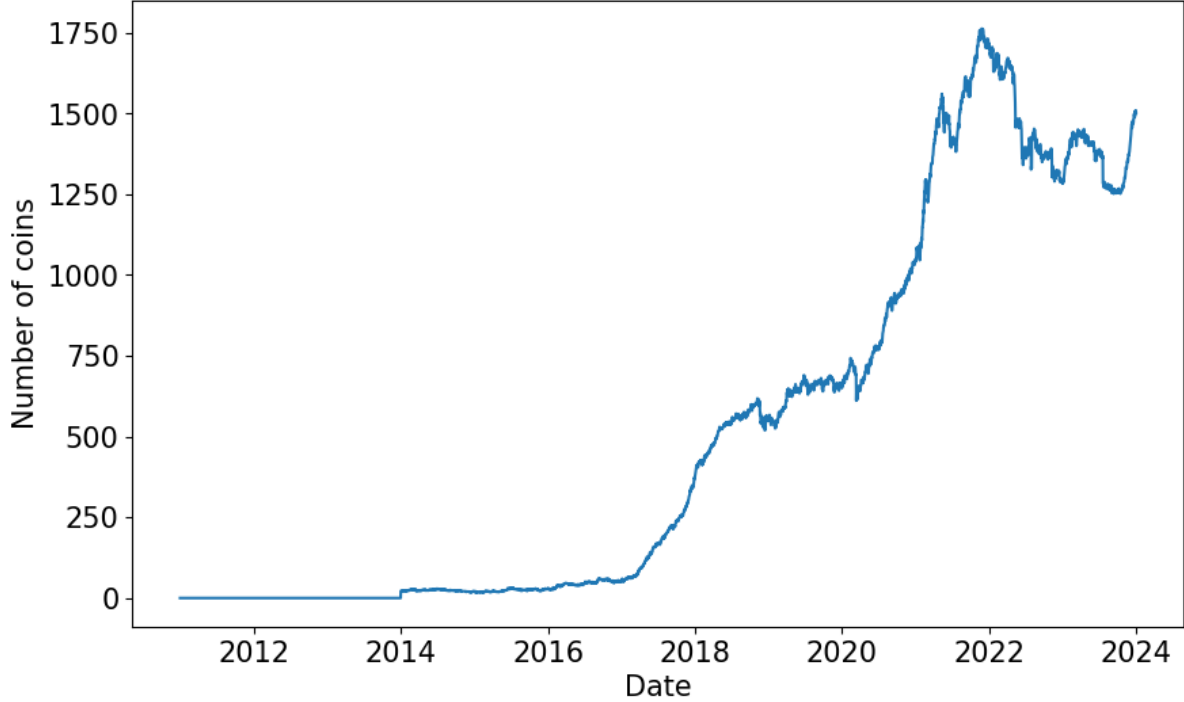
2. Data

We collect cryptocurrency data from Coinmarketcap.com, a widely used data source in previous studies (Liu and Tsyvinski, 2021, Liu et al., 2022, Shen et al., 2020). It provides daily price, volume, and market capitalization data in US dollars aggregated from hundreds of cryptocurrency exchanges. Following Fieberg et al. (2023) and Liu et al. (2022), we apply filters to clean our data. First, we exclude stablecoins in our analysis because stablecoins are pegged to external financial assets or designed to have zero returns, with a distinct nature from other cryptocurrencies. Second, we only consider cryptocurrencies with a market capitalization surpassing \$1 million daily, hence, the number of coins naturally differs across our sample. Figure 1 shows the number of cryptocurrencies satisfying the filtering criteria from 2011 to 2023. In the initial years, the cryptocurrency market was dominated by Bitcoin. There is a notable increase in coins in 2014, with more than 20 cryptocurrencies meeting the criteria. In light of the diversification of the market, we select our sample period from 1st January 2014 to 31st December 2023, encompassing a total of 3,192 coins included in our data. Panel A of Table 1 provides the summary statistics of our sample by year. The average daily market capitalization is \$548.69 million and the average daily trading volume is \$41.14 million. The number of coins in our sample ranges from 53 in 2014 to more than 1,000 in 2020. As of 31st December 2023, the total market capitalization and daily trading volume rise to \$1,604.49 billion and \$49.25 billion, respectively.

To understand the time series momentum (TSMOM) effect in the cryptocurrency market, we

Figure 1: The number of coins with a market capitalisation surpassing \$1 million

This figure illustrates the number of coins after filtering by excluding stablecoins and coins with a market capitalisation below \$1 million daily from January 2011 to December 2023. Since the number of coins in our sample substantially expands in 2014, we select our sample period starting from 2014.



first construct a volume-weighted market return of all coins instead of using the capital-weighted return as in (Liu and Tsyvinski, 2021) because cryptocurrency market capitalization could be affected by modifications in the cryptocurrency system, which are not foreseen nor expected by traders. For example, manipulating the coin supply can impact market capitalization.³ An intentional reduction in a coin’s supply to create scarcity can lead to an elevation in its price and overall market value. Moreover, we find that some cryptocurrencies, such as Peercoin and Namecoin, have a market capitalization of more than \$1 million, but a daily trading volume lower than \$10,000. This suggests that trading volume may serve as an indicator of a cryptocurrency’s genuine popularity compared to market capitalization. Some studies (Aloosh and Li, 2024, Cong et al., 2023) demonstrate that cryptocurrency exchanges suspect wash trading, therefore, we conduct

³<https://coinswitch.co/switch/crypto/how-does-market-cap-affect-crypto-prices/>

Table 1: Summary statistics

The two tables tabulate the summary statistics of our sample data. Panel A provides the number of coins and the mean and median of market capitalisation and daily trading volume in USD by year. We only consider coins with a market capitalisation above \$1 million daily sampling from January 2014 to December 2023. Panel B compares the summary statistics of the daily return of the volume-weighted market return (VW market), capital-weighted market return (CW market), equally-weighted market return (EQ market), Bitcoin, Ethereum, and XRP.

Panel A: Sample data					
Year	Count	Market Cap (mil)		Volume (thous)	
		Mean	Median	Mean	Median
2014	53	316.58	5.75	1,511.97	38.87
2015	47	190.28	4.91	1,646.56	18.07
2016	97	233.90	4.48	2,689.01	27.72
2017	397	585.20	13.69	21,954.00	211.30
2018	828	603.10	15.53	29,611.12	252.16
2019	973	324.33	6.27	57,091.89	226.61
2020	1326	397.53	7.45	94,613.27	344.72
2021	2325	1,234.89	14.92	113,737.73	702.61
2022	2142	796.76	11.26	56,155.63	494.34
2023	2025	805.58	12.98	32,349.96	426.16
Full	3192	548.69	8.90	41,140.23	262.91
Panel B: Volume-weighted market return and major cryptocurrencies					
	Mean	Median	STD	Skewness	Kurtosis
VW market	0.62%	0.56%	0.05	-0.97	11.26
CW market	0.21%	0.23%	0.04	-0.25	16.18
EQ market	0.33%	0.51%	0.04	-1.21	10.86
Bitcoin	0.11%	0.11%	0.04	-0.76	10.95
Ethereum	0.26%	0.04%	0.06	0.01	8.74
XRP	0.09%	-0.17%	0.06	1.81	35.18

robustness checks with Coin Metrics database, which reports the volume of trusted exchanges.⁴

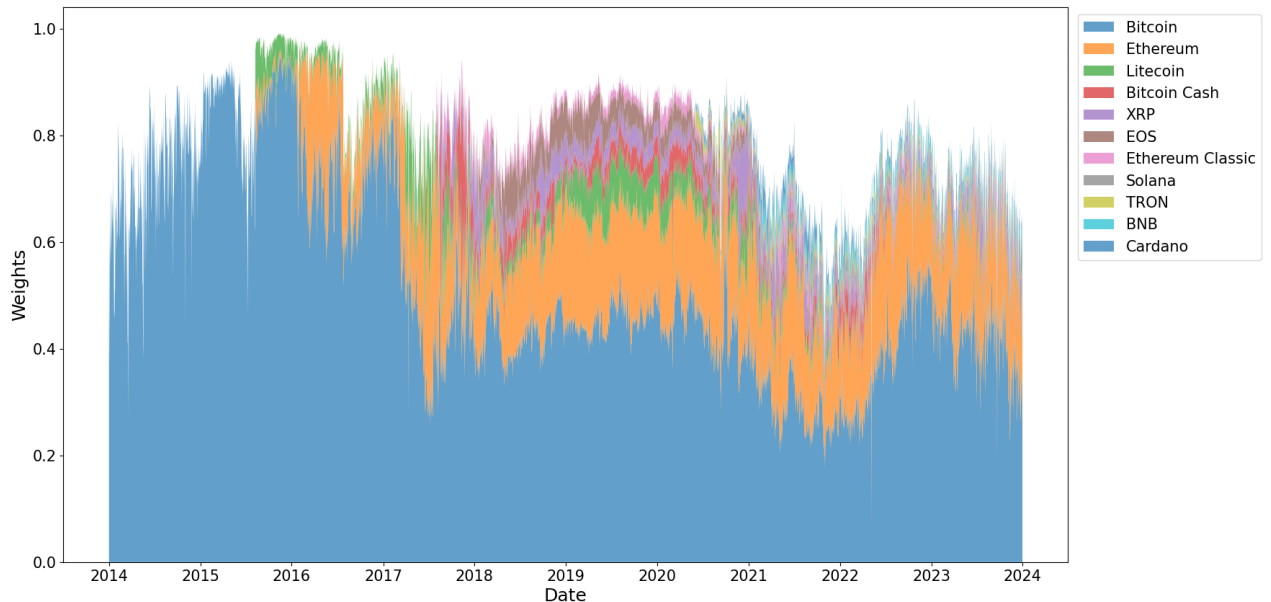
The volume-weighted market return is calculated by summing up each coin’s return weighted by the daily trading volume. Figure 2 shows the market dominance (volume weights) of the top 11 cryptocurrencies ranked by average trading volume of all time. Bitcoin prevails in the

⁴The trusted exchanges are selected based on public information of exchange, such as financial statements and disclosure, to ensure the data quality. Detailed selection criteria: <https://coinmetrics.io/special-insights/trusted-exchange-framework-2-1/>

cryptocurrency market occupying 52.62% of total trading volume on average, followed by Ethereum with 16.82%. In 2018, we can observe a more diversified cryptocurrency market, when more cryptocurrencies demonstrate an elevated proportion of trading volume, and concurrently, Bitcoin's market dominance drops drastically to around 50% of the market. Therefore, we also analyse the periods before and after 2018. Panel B of Table 1 provides the summary statistics of the volume-weighted (VW) market return, capital-weighted (CW) market return, equally-weighted (EQ) market return, and three major coins' daily returns including Bitcoin, Ethereum, and XRP. The average daily VW market return is 0.62%, which is higher than the average daily return of the CW market (0.21%), EQ market (0.33%), Bitcoin (0.11%), Ethereum (0.26%), and XRP (0.09%). The daily standard deviation of the market returns is 0.05 (VW) and 0.04 (CW and EQ), which is close to that of Bitcoin (0.04), Ethereum (0.06), and XRP (0.06). The market returns exhibit a negative skewness with a large kurtosis, indicating the distribution of the market return tends to have heavy tails or outliers.

Figure 2: Major cryptocurrencies' volume dominance in the cryptocurrency market

This graph shows the proportion of the volume-weighted index occupied by the top 11 cryptocurrencies, whose volume weights are on average above 10% across the whole sample period.



3. Time series momentum of the volume-weighted market return

We start by testing the daily TSMOM for the volume-weighted market return.⁵ Specifically, we regress the current future volume-weighted market return on the past cumulative volume-weighted market return as follows:

$$R_{VW_t} = \alpha + R_{VW_{t-1:t-h}} + e \quad (1)$$

where $R_{VW_{t-1:t-h}}$ is the cumulative future volume-weighted market return aggregated from various time horizons spanning from the past 1-day to the past 2-week, R_{VW_t} is current volume-weighted market return, α and e are the intercept and residual of the regression.⁶

Table 2 presents evidence of a strong TSMOM effect in volume-weighted market return. The results indicate that the current market return can be positively predicted by the past cumulative market return at a 1% significance level for most of the time horizons (from 3-day to 2-week) and across all sampling periods. We conjecture that the noise in a 1- or 2-day time horizon might affect the results and lead to its insignificance. The highest coefficients are 0.042, 0.041, and 0.041 for the sample period of 2014 to 2023, 2014 to 2017, and 2018 to 2023, respectively. We can observe stronger TSMOM before 2018, which may result from the cryptocurrency market having become more diverse since 2018 when a small group of "big coins" accounted for a smaller portion of the market. The diversification spreads out investment across a broader range of assets, which can help mitigate the impact of price movements on any single coin.

Following Liu and Tsyvinski (2021), we further test the TSMOM magnitude of the cryptocurrency market by grouping the past cumulative volume-weighted market returns into terciles and

⁵Section 2 states the volume-weighted market return calculation.

⁶In this paper, if we do not specify, regression refers to Ordinary Least Square regression, with t-statistics adjusted using Newey-West's heteroskedasticity and autocorrelation consistent estimator and $[T^{1/4}]$ lags (Newey and West, 1994), where T is the sample size.

Table 2: Time-series momentum of the volume-weighted market return

This table presents the TSMOM results for the volume-weighted market return by regressing the current market return on the past cumulative market return among different time horizons. We show the results of cumulative market return from 1-day to 2-week with the full sample period, from 2014 to 2023, and subsamples of 2014 to 2017 and 2018 to 2023. The t-statistics are shown in parentheses.

2014-01-01 - 2023-12-31														
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.015	0.028*	0.039***	0.042***	0.035***	0.039***	0.033***	0.029***	0.027***	0.028***	0.027***	0.025***	0.024***	0.024***
	(0.51)	(1.66)	(2.72)	(3.63)	(3.62)	(4.76)	(4.23)	(4.17)	(4.16)	(4.73)	(4.62)	(4.44)	(4.46)	(4.70)
Adj. R^2	-0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
2014-01-01 - 2017-12-31														
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.031	0.014	0.039*	0.041**	0.038**	0.047***	0.039***	0.037***	0.035***	0.037***	0.036***	0.033***	0.031***	0.033***
	(0.85)	(0.52)	(1.88)	(2.33)	(2.49)	(3.61)	(3.27)	(3.38)	(3.47)	(3.98)	(4.02)	(4.02)	(3.96)	(4.46)
Adj. R^2	0.00	-0.00	0.00	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02
2018-01-01 - 2023-12-31														
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.004	0.036*	0.037*	0.041***	0.033**	0.033***	0.028***	0.023***	0.021**	0.022***	0.019***	0.018**	0.017***	0.016**
	(0.08)	(1.69)	(1.94)	(2.69)	(2.56)	(3.15)	(2.76)	(2.61)	(2.54)	(2.85)	(2.69)	(2.49)	(2.59)	(2.54)
Adj. R^2	-0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.00

*** p<0.01, ** p<0.05, * p<0.1

Table 3: Time-series momentum by groups

Following Liu and Tsyvinski (2021), the two tables estimate the time-series momentum magnitude of the volume-weighted market return. Panel A presents the results by grouping volume-weighted market past cumulative returns into terciles labelled as Low, Middle, and High, and evaluating the subsequent day's return. Panel B groups the volume-weighted cumulative returns by the cutoffs calculated by the first two years' data (from 2014 to 2015), and evaluates the out-of-sample (from 2016 to 2023) TSMOM magnitude of the cryptocurrencies market. The results are in percentage expression and the t-statistics are in brackets. Diff refers to the difference in mean return between the High group and the Low group. Weekly High Minus Low (HML) is the difference between the total weekly return of the High group and that of the Low group.

Panel A: Time-series momentum by groups (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-3.66*** (-31.39)	0.41*** (2.74)	0.20 (1.25)	0.17 (1.08)	0.21 (1.39)	0.13 (0.84)	0.12 (0.79)	0.11 (0.73)	0.13 (0.87)
Middle	0.59*** (31.24)	0.65*** (5.59)	0.49*** (4.77)	0.52*** (5.18)	0.44*** (4.19)	0.52*** (4.85)	0.36*** (3.27)	0.46*** (4.26)	0.46*** (4.47)
High	4.93*** (54.25)	0.80*** (6.18)	1.16*** (8.89)	1.15*** (8.72)	1.20*** (8.90)	1.19*** (9.02)	1.36*** (10.23)	1.29*** (9.80)	1.26*** (9.12)
Diff	8.59	0.39	0.96	0.99	0.98	1.06	1.24	1.17	1.13
Weekly HML	20.03*** (33.13)	0.90* (1.92)	2.24*** (4.42)	2.29*** (4.45)	2.29*** (4.25)	2.48*** (4.47)	2.89*** (5.39)	2.74*** (5.07)	2.64*** (4.76)
Panel B: Time-series momentum by groups with the first 2-year cutoffs (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-4.14*** (-27.83)	0.54*** (2.78)	0.23 (1.10)	0.24 (1.06)	0.12 (0.55)	0.22 (0.95)	0.12 (0.49)	0.18 (0.71)	0.13 (0.54)
Middle	0.29*** (14.35)	0.75*** (5.68)	0.48*** (3.66)	0.37*** (2.84)	0.53*** (4.02)	0.55*** (3.99)	0.33*** (2.66)	0.20 (1.62)	0.40*** (3.10)
High	4.57*** (48.73)	0.87*** (6.57)	1.21*** (10.08)	1.23*** (10.50)	1.16*** (9.83)	1.08*** (9.45)	1.23*** (10.46)	1.26*** (10.98)	1.14*** (9.90)
Diff	8.71	0.33	0.98	0.98	1.03	0.86	1.11	1.09	1.01
Weekly HML	20.91*** (30.27)	1.36** (2.45)	3.29*** (5.39)	3.59*** (5.82)	3.71*** (5.70)	3.40*** (5.11)	4.15*** (6.33)	4.23*** (6.48)	4.07*** (6.20)

*** p<0.01, ** p<0.05, * p<0.1

calculating the following one day’s average market return of each group. In Panel A of Table 3, the average daily return of terciles is -3.66% (Low), 0.59% (Middle), and 4.93% (High). The top tercile outperforms the bottom tercile for 1-day to 2-week time horizons,⁷ with all positive differences between High and Low group mean returns (Diff). Moreover, we determine the weekly High Minus Low return (Weekly HML) by subtracting the aggregate return of days labelled as Low from the total return of days categorised as High within a week. We find significant positive Weekly HML among all time horizons, peaking at an average weekly return of 2.89%.

Furthermore, in Panel B of Table 3, we examine the out-of-sample TSMOM magnitude. We categorise the market returns by the tercile cutoffs determined by the first 2-year volume-weighted cumulative returns (from 2014 to 2015) and assess the subsequent years’ returns (from 2016 to 2023) within each group. The average daily return of each group is -4.14% (Low), 0.29% (Middle), and 4.57% (High). We can observe a significant TSMOM effect as in Panel A, confirming that higher past returns can generate greater future returns. The highest Diff and Weekly HML are 1.11% and 4.23%, respectively. To conclude, our study reveals the presence of the strong daily TSMOM effect in the cryptocurrency market ranging from 1-day to 2-week time horizons.

4. Time series momentum strategy

4.1. Portfolio construction

In this section, we construct the volume-weighted Winner Minus Loser (WML) portfolio based on the TSMOM strategy. First, based on the past formation days’ cumulative returns, we classify the past positive ones as winners while the negative ones as losers. Second, we apply volume weighting within the winner (loser) portfolios, where each coin is weighted by its daily trading

⁷For simplicity, we present results for 1-day to 7-day and 14-day time horizons. Full results spanning from 1-day to 2-week are qualitatively the same, showing strong TSMOM, and are available upon request.

volume divided by the total daily trading volume of the winner (loser) portfolios. We create a volume-weighted long-short portfolio by buying winners and selling losers and holding the portfolio for 1-day. The portfolio rebalances daily and is dollar-neutral (zero net investment). Moreover, we also create the WML portfolios weighting by market capitalization, inverse volatility, and equal proportion to make comparisons following Lim et al. (2018). We compare the performance of WML portfolios with different weighting schemes using the average return and the Sharpe ratio (Sharpe, 1966), which measures units of excess return per unit of risk and is defined as follows:

$$SR_p = \frac{\overline{R_p} - R_f}{\sigma_p} \quad (2)$$

where SR_p is the Sharpe ratio of portfolio p , $\overline{R_p}$ is the mean return of portfolio p , σ_p is the standard deviation of portfolio p 's excess return, and R_f is the risk-free rate.⁸

Panel A of Table 4 provides the average return of the WML portfolio for a set of formation periods, 1-day, 7-day, and 14-day, with a 1-day holding period.⁹ We find that the volume-weighted WML (VW-WML) portfolio outperforms all the other weighting schemes in terms of WML portfolio daily mean return. The VW-WML portfolios have the highest positive daily average returns of 0.94% (1-day), 0.87% (7-day), and 0.67% (14-day), more than twice the capital-weighted WML portfolio (CW-WML) average returns, with 0.33% (1-day), 0.28% (7-day), and 0.24% (14-day). The annualised Sharpe ratios of the VW-WML portfolio are 2.17 (1-day), 2.66 (7-day), and 1.86 (14-day), indicating high performance despite the significant volatility in the cryptocurrency market.

Figure 3 presents the cumulative WML portfolio returns in comparison to the volume-weighted

⁸We use the 1-Month Treasury Bill rate obtained from Wharton Research Data Services as the risk-free rate.

⁹For simplicity, we only present results for 1-day, 7-day, and 14-day time horizons. Full results spanning from 1-day to 2-week are qualitatively the same, showing strong TSMOM, and are available upon request.

Table 4: Time series momentum strategy Winner Minus Loser (WML) long-short portfolio

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. The WML strategy longs the coins with positive past returns and shorts the coins with negative past returns. Within the long and short portfolios, we weight coins by trading volume, market capitalisation, equal proportion, and inverse volatility. We show the results of lookback periods of 1-day, 7-day, and 14-day. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1% (Liu and Tsyvinski, 2021). The Sharpe ratios are annualised by 365 days.

Panel A: WML portfolio performance									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.19*** (9.54)	0.25** (2.29)	0.94*** (6.91)	1.09*** (10.20)	0.21** (2.44)	0.87*** (8.45)	1.04*** (9.30)	0.37*** (4.06)	0.67*** (5.94)
Sharpe ratio			2.17			2.66			1.86
Capital-weighted average return (daily, percentage)	0.48*** (4.15)	0.15* (1.79)	0.33*** (2.87)	0.34*** (3.46)	0.06 (0.70)	0.28*** (2.80)	0.37*** (3.83)	0.13* (1.76)	0.24*** (2.65)
Sharpe ratio			0.89			0.87			0.82
Equally-weighted average return (daily, percentage)	-0.76*** (-9.82)	0.43*** (6.00)	-1.19*** (-21.87)	-0.70*** (-9.12)	0.20*** (2.93)	-0.91*** (-16.60)	-0.62*** (-8.05)	0.18** (2.57)	-0.80*** (-14.53)
Sharpe ratio			-6.94			-5.27			-4.62
Volatility-weighted average return (daily, percentage)	-	-	-	-0.12* (-1.86)	-0.06 (-0.87)	-0.07 (-1.13)	-0.17** (-2.38)	-0.01 (-0.18)	-0.15** (-2.53)
Sharpe ratio			-			-0.38			-0.82
Panel B: WML portfolio performance considering transaction cost of 10 bps									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.09*** (8.74)	0.15 (1.37)	0.74*** (5.44)	0.99*** (9.26)	0.11 (1.30)	0.67*** (6.52)	0.94*** (8.40)	0.27*** (2.97)	0.47*** (4.15)
Sharpe ratio			1.71			2.05			1.3
Capital-weighted average return (daily, percentage)	0.38*** (3.29)	0.05 (0.61)	0.13 (1.13)	0.24** (2.44)	-0.04 (-0.50)	0.08 (0.80)	0.27*** (2.79)	0.03 (0.41)	0.04 (0.43)
Sharpe ratio			0.35			0.24			0.12
Equally-weighted average return (daily, percentage)	-0.86*** (-11.10)	0.33*** (4.59)	-1.39*** (-25.55)	-0.80*** (-10.41)	0.1 (1.48)	-1.11*** (-20.26)	-0.72*** (-9.34)	0.08 (1.11)	-1.00*** (-18.17)
Sharpe ratio			-8.1			-6.43			-5.77
Volatility-weighted average return (daily, percentage)	-	-	-	-0.22*** (-3.39)	-0.16** (-2.44)	-0.27*** (-4.58)	-0.27*** (-3.81)	-0.11* (-1.76)	-0.35*** (-5.82)
Sharpe ratio			-			-1.47			-1.86

*** p<0.01, ** p<0.05, * p<0.1

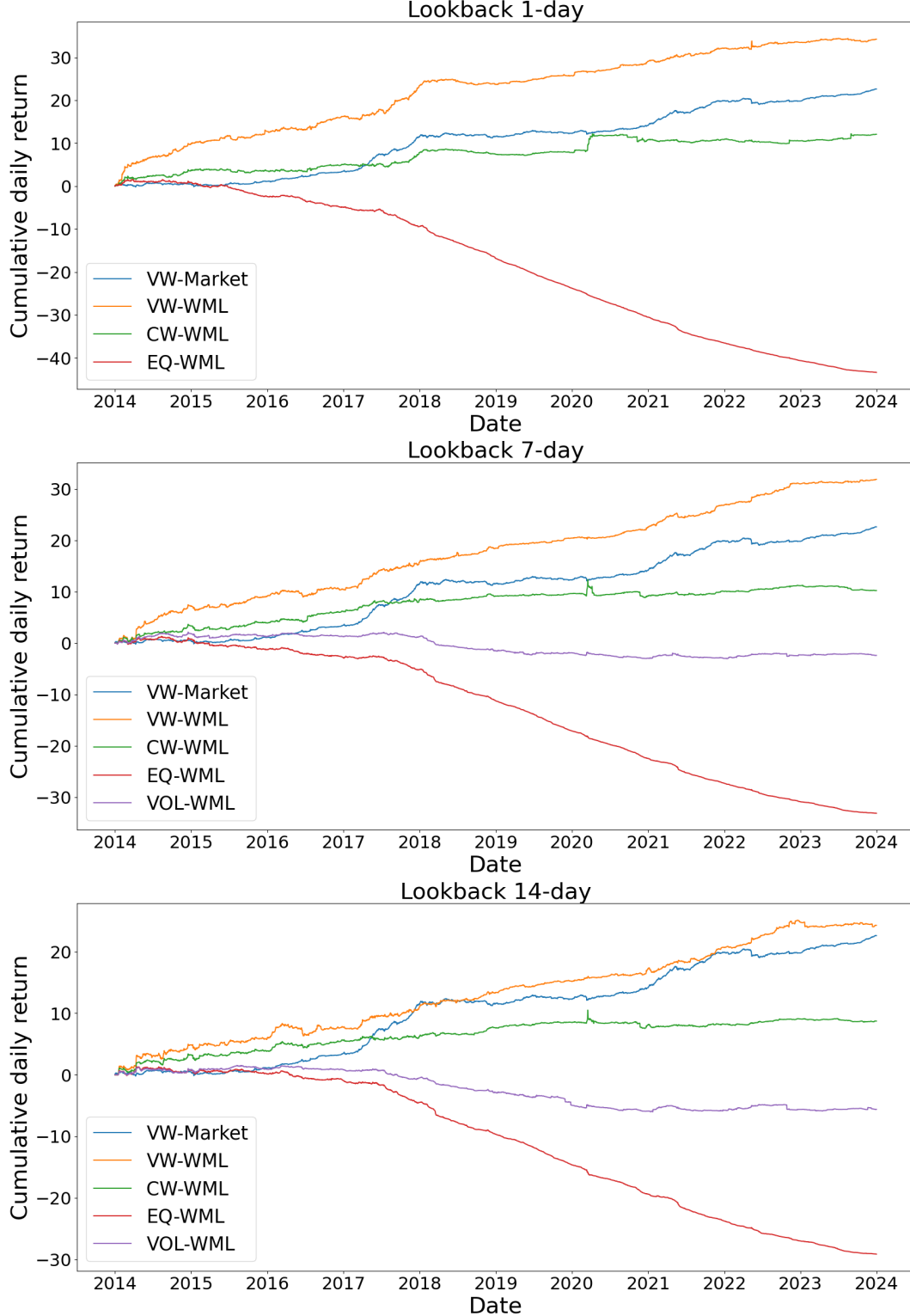
market return computed in Section 2. Compared to all the other weighting schemes, the VW-TSMOM outperforms the market of all time. Although Liu and Tsyvinski (2021) demonstrate a strong TSMOM effect in capital-weighted market return, we find that the VW-WML can generate average returns greater than capital-weighted WML portfolios. The majority of market capitalization flows in the small group of "big" cryptocurrencies, such as Bitcoin and Ethereum. Hence, market capitalization cannot measure the trading activities of the coin market, leading to underperforming WML portfolios.

Following Lim et al. (2018), we also compare to equally-weighted WML portfolio (EQ-WML) and inverse volatility-weighted WML portfolio (VOL-WML). It is not surprising that the mean return of the two weighting schemes are all negative. The equally-weighted WML portfolios apply the same weights to all coins, which undoubtedly does not work in the cryptocurrency market because some coins do not have enough investors' attention nor sufficient liquidity, with significantly less trading volume and market capitalization. Additionally, the efficacy of inverse volatility-weighted WML portfolios is also questioned because cryptocurrency market investors are considered more willing to take higher risks as compensation for greater returns compared to investors in other traditional financial assets. We find that with weighting by the daily trading volume, VW-WML portfolios can be the most profitable compared to other weighting schemes.

Table 5 gives the ratio of days on which either buying or selling a cryptocurrency within the VW-WML portfolio for the major 11 coins, reported in Section 2, as well as the average and median proportions of all coins. Cryptocurrencies in the VW-WML portfolio are on average 48.50% (1-day formation), 43.08% (7-day formation), and 40.97% (14-day formation) of time in a long position, and 51.50% (1-day formation), 56.92% (7-day formation), and 59.03% (14-day formation) of time in a short position. The percentages of long and short positions varied among the 11 top coins. Bitcoin, which dominated the coin market, is more traded in a long position, with 52.42% (1-

Figure 3: Winner Minus Loser (WML) long-short portfolio cumulative return over time

The three figures plot the WML portfolio cumulative returns of different weighting schemes, including volume (VW-WML), market capitalisation (CW-WML), equal proportion (EQ-WML), and inverse volatility (VOL-WML) weighting. We also include the volume-weighted cryptocurrency market (VW-Market) cumulative return calculated in Section 2 as comparisons to WML portfolio returns.



day formation), 53.5% (7-day formation), and 54.92% (14-day formation) of days in the winner portfolio. Table 5 demonstrates that although a small group of "big" coins, such as Bitcoin and Ethereum, have higher trading volume, there is no dominant coin of the WML portfolios.

Table 5: Long and short percentages in WML portfolio (percentage)

This table shows the percentage of the number of times a coin is long or short in the WML portfolios with a lookback period of 1-day, 7-day, and 14-day. We present the results of the average of all coins and the top 11 coins reported in Section 2.

Lookback Name	1-day		7-day		14-day	
	Long	Short	Long	Short	Long	Short
Bitcoin	52.42	47.58	53.50	46.50	54.92	45.08
Ethereum	50.59	49.41	53.01	46.99	54.21	45.79
Litecoin	49.58	50.42	50.32	49.68	49.09	50.91
Bitcoin Cash	48.74	51.26	47.58	52.42	45.11	54.89
XRP	47.49	52.51	44.88	55.12	43.05	56.95
EOS	49.66	50.34	49.75	50.25	47.48	52.52
Ethereum Classic	49.23	50.77	47.08	52.92	46.32	53.68
TRON	50.34	49.66	52.04	47.96	52.70	47.30
Solana	52.39	47.61	52.79	47.21	52.55	47.45
BNB	52.09	47.91	53.63	46.37	55.07	44.93
Cardano	50.00	50.00	48.37	51.63	49.58	50.42
Mean of all coins	48.50	51.50	43.08	56.92	40.97	59.03
Median of all coins	47.65	52.35	43.75	56.25	41.88	58.12
*** p<0.01, ** p<0.05, * p<0.1						

4.2. WML portfolios after controlling for cryptocurrency factors

We choose the VW-WML portfolio with 1-day, 7-day, and 14-day lookback periods and a 1-day holding period to examine whether the three-factor model suggested in Liu et al. (2022) can explain the portfolio's excess return.¹⁰ First, we construct the cryptocurrency excess market return (CMKT) by calculating capital-weighted market return minus the risk-free rate.¹¹ Second, to create the cryptocurrency size factor, CSMB, we categorise coins into the top 30% (Big), the

¹⁰For simplicity, we only present results for 1-day, 7-day, and 14-day time horizons. Full results spanning from 1-day to 2-week are available upon request.

¹¹The risk-free rate is calculated from the 1-Month Treasury Bill rate downloaded from Wharton Research Data Services.

middle 40%, and the bottom 30% (Small) by coins' daily market capitalization and calculate CSMB by the return of the small-size portfolio minus which of the big-size portfolio. Third, to create the momentum factor, CMOM, we sort cryptocurrencies into two portfolios by coin size. Then, we construct three momentum portfolios within two sizes based on past days' returns with the top 30%, the middle 40%, and the bottom 30%. The CMOM factor is computed as follows:

$$CMOM = \frac{1}{2}(Small\ High + Big\ High) - \frac{1}{2}(Small\ Low + Big\ Low) \quad (3)$$

Table 6 presents the regression results of WML portfolio daily excess return by controlling standard common risk factors, including CMKT, CSMB, and CMOM,¹² using the following models:

$$R_i - R_f = \alpha_i + \beta_{1i}(CMKT - R_f) + \epsilon_i \quad (4)$$

$$R_i - R_f = \alpha_i + \beta_{1i}(CMKT - R_f) + \beta_{2i}CSMB + \epsilon_i \quad (5)$$

$$R_i - R_f = \alpha_i + \beta_{1i}(CMKT - R_f) + \beta_{2i}CMOM + \epsilon_i \quad (6)$$

$$R_i - R_f = \alpha_i + \beta_{1i}(CMKT - R_f) + \beta_{2i}CSMB + \beta_{3i}CMOM + \epsilon_i \quad (7)$$

where R_i is WML portfolio return, and R_f is risk-free rate. The results in Table 6 show that the alphas when controlling all common cryptocurrency factors for VW-WML and CW-WML portfolios are significantly positive with values of 0.022 and 0.011, respectively. The adjusted R-squared is small with the highest value of 0.1, meaning that the VW-WML portfolio return cannot be well explained by three factors, which conflicts with the findings in Liu et al. (2022).

¹²Appendix C.2 provides the results for the sub-sample periods before and after 2018, which results are similar to the full sample period.

Table 6: Winner Minus Loser (WML) portfolios after controlling for cryptocurrency factors

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.009*** (7.08)	0.009*** (7.13)	0.021*** (10.38)	0.022*** (10.45)	0.003*** (2.80)	0.003*** (2.98)	0.011*** (7.53)	-0.012*** (-18.46)	-0.012*** (-18.73)	0.000 (-0.47)	0.000 (-0.28)	-	-	-	-
CMKT-Rf	0.03 (0.53)	0.02 (0.31)	0.05 (1.04)	0.03 (0.62)	0.14* (1.88)	0.12* (1.68)	0.15** (2.13)	0.00 (-0.16)	0.00 (0.12)	0.02* (1.74)	0.02 (1.45)				
CSMB	-0.08* (-1.70)	-0.08* (-1.70)	-0.13*** (-3.06)	-0.13*** (-3.06)	-0.12*** (-2.68)	-0.12*** (-2.68)	-0.15*** (-3.46)	0.03 (1.47)	0.03 (1.47)	-0.02 (-1.49)	-0.02 (-1.49)				
CMOM			0.76*** (9.39)	0.77*** (9.53)	0.77*** (9.53)	0.77*** (9.53)	0.48*** (8.62)	0.50*** (9.03)	0.50*** (9.03)	0.73*** (33.52)	0.74*** (33.87)				
Adj. R^2	0.00	0.00	0.10	0.10	0.01	0.01	0.06	0.07	0.00	0.59	0.59				
7-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.009*** (7.41)	0.009*** (7.42)	0.016*** (10.71)	0.017*** (10.70)	0.003*** (2.65)	0.003*** (2.64)	0.008*** (7.04)	-0.009*** (-14.34)	-0.009*** (-14.42)	0.000 (0.20)	0.000 (0.02)	0.000 (-0.44)	0.000 (-0.47)	0.005*** (7.62)	0.005*** (7.39)
CMKT-Rf	0.05 (1.48)	0.04 (1.24)	0.08*** (2.77)	0.07** (2.37)	0.02 (0.34)	0.01 (0.17)	0.04 (0.88)	-0.04* (-1.81)	-0.03 (-1.44)	0.00 (-0.08)	0.00 (0.18)	-0.21*** (-5.30)	-0.21*** (-5.08)	-0.19*** (-5.70)	-0.19*** (-5.53)
CSMB	-0.04 (-1.09)	-0.04 (-1.09)	-0.06* (-1.71)	-0.06* (-1.71)	-0.05 (-1.06)	-0.05 (-1.06)	-0.07 (-1.37)	0.05** (2.04)	0.05** (2.04)	0.02 (0.34)	0.02 (0.34)			0.00 (-0.19)	0.00 (-0.19)
CMOM			0.61*** (10.08)	0.62*** (10.09)	0.62*** (10.09)	0.62*** (10.09)	0.45*** (8.62)	0.45*** (8.66)	0.45*** (8.66)	0.73*** (34.73)	0.72*** (34.57)			0.44*** (16.37)	0.44*** (16.27)
Adj. R^2	0.00	0.00	0.10	0.10	0.00	0.00	0.06	0.06	0.01	0.51	0.51	0.06	0.06	0.23	0.23
14-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (5.26)	0.007*** (5.48)	0.013*** (8.50)	0.014*** (8.72)	0.002** (2.44)	0.003*** (2.80)	0.007*** (6.37)	-0.008*** (-12.18)	-0.008*** (-12.05)	0.000 (0.18)	0.000 (0.25)	-0.001* (-1.87)	-0.001* (-1.69)	0.004*** (5.81)	0.004*** (5.96)
CMKT-Rf	0.09*** (2.60)	0.08** (2.18)	0.12*** (3.28)	0.10*** (2.80)	0.11*** (2.70)	0.09** (2.36)	0.12*** (3.12)	-0.02 (-0.80)	-0.02 (-0.84)	0.01 (0.74)	0.01 (0.58)	-0.17*** (-4.29)	-0.18*** (-4.32)	-0.16*** (-4.24)	-0.16*** (-4.29)
CSMB	-0.09** (-2.33)	-0.09** (-2.33)	-0.09*** (-2.82)	-0.09*** (-2.82)	-0.12*** (-3.09)	-0.12*** (-3.09)	-0.12*** (-3.30)	-0.01 (-0.41)	-0.01 (-0.41)	-0.03 (-1.06)	-0.03 (-1.06)			-0.04** (-2.21)	-0.04** (-2.21)
CMOM			0.66*** (9.27)	0.66*** (9.32)	0.66*** (9.32)	0.66*** (9.32)	0.45*** (8.71)	0.45*** (8.79)	0.45*** (8.79)	0.76*** (27.14)	0.76*** (27.21)			0.51*** (18.43)	0.51*** (18.52)
Adj. R^2	0.00	0.01	0.10	0.10	0.01	0.01	0.08	0.08	0.00	0.53	0.53	0.04	0.04	0.24	0.24
*** p<0.01, ** p<0.05, * p<0.1															

*** p<0.01, ** p<0.05, * p<0.1

Table 7: Winner Minus Loser (WML) portfolios with transaction costs after controlling for cryptocurrency factors

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.007*** (5.55)	0.007*** (5.63)	0.019*** (9.40)	0.020*** (9.50)	0.001 (0.92)	0.001 (1.20)	0.009*** (6.11)	0.009*** (6.36)	-0.014*** (-21.55)	-0.014*** (-21.84)	-0.002*** (-4.15)	-0.002*** (-3.96)	-	-	-	-
CMKT-Rf	0.03 (0.53)	0.02 (0.31)	0.05 (1.04)	0.03 (0.62)	0.14* (1.88)	0.12* (1.68)	0.15** (2.13)	0.13* (1.85)	0.00 (-0.16)	0.00 (0.12)	0.02* (1.74)	0.02 (1.45)				
CSMB		-0.08* (-1.70)		-0.13*** (-3.06)		-0.12*** (-2.68)		-0.15*** (-3.46)		0.03 (1.47)		-0.02 (-1.49)				
CMOM			0.76*** (9.39)	0.77*** (9.53)			0.48*** (8.62)	0.50*** (9.03)			0.73*** (33.52)	0.74*** (33.87)				
Adj. R^2	0.00	0.00	0.10	0.10	0.01	0.01	0.06	0.07	0.00	0.00	0.59	0.59				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.007*** (5.68)	0.007*** (5.71)	0.014*** (9.40)	0.015*** (9.41)	0.001 (0.70)	0.001 (0.81)	0.006*** (5.35)	0.007*** (5.26)	-0.011*** (-17.51)	-0.011*** (-17.56)	-0.002*** (-3.83)	-0.002*** (-3.89)	-0.002*** (-3.61)	-0.002*** (-3.57)	0.003*** (4.74)	0.003*** (4.61)
CMKT-Rf	0.05 (1.48)	0.04 (1.24)	0.08*** (2.77)	0.07** (2.37)	0.02 (0.34)	0.01 (0.17)	0.04 (0.88)	0.03 (0.71)	-0.04* (-1.81)	-0.03 (-1.44)	0.00 (-0.08)	0.00 (0.18)	-0.21*** (-5.30)	-0.21*** (-5.08)	-0.19*** (-5.70)	-0.19*** (-5.53)
CSMB		-0.04 (-1.09)		-0.06* (-1.71)		-0.05 (-1.06)		-0.07 (-1.37)		0.05** (2.04)		0.02 (1.47)		0.01 (0.34)		0.00 (-0.19)
CMOM			0.61*** (10.08)	0.62*** (10.09)			0.45*** (8.62)	0.45*** (8.66)			0.73*** (34.73)	0.72*** (34.57)	0.44*** (16.37)	0.44*** (16.37)		0.44*** (16.27)
Adj. R^2	0.00	0.00	0.10	0.10	0.00	0.00	0.06	0.06	0.00	0.01	0.51	0.51	0.06	0.06	0.23	0.23
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.004*** (3.62)	0.005*** (3.84)	0.011*** (7.25)	0.012*** (7.46)	0 (0.15)	0.001 (0.56)	0.005*** (4.55)	0.005*** (4.86)	-0.010*** (-15.23)	-0.010*** (-15.08)	-0.002*** (-3.18)	-0.002*** (-3.14)	-0.003*** (-4.90)	-0.003*** (-4.69)	0.002*** (3.07)	0.002*** (3.22)
CMKT-Rf	0.09*** (2.60)	0.08** (2.18)	0.12*** (3.28)	0.10*** (2.80)	0.11*** (2.70)	0.09** (2.36)	0.12*** (3.12)	0.10*** (2.80)	-0.02 (-0.80)	-0.02 (-0.84)	0.01 (0.74)	0.01 (0.58)	-0.17*** (-4.29)	-0.18*** (-4.32)	-0.16*** (-4.24)	-0.16*** (-4.29)
CSMB		-0.09** (-2.33)		-0.09*** (-2.82)		-0.12*** (-3.09)		-0.12*** (-3.30)		-0.01 (-0.41)		-0.01 (-1.06)		-0.03 (-1.64)		-0.04** (-2.21)
CMOM			0.66*** (9.27)	0.66*** (9.32)			0.45*** (8.71)	0.45*** (8.79)			0.70*** (27.14)	0.70*** (27.21)	0.51*** (18.43)	0.51*** (18.43)		0.51*** (18.52)
Adj. R^2	0.00	0.01	0.10	0.10	0.01	0.01	0.08	0.08	0.00	0.00	0.53	0.53	0.04	0.04	0.24	0.24

*** p<0.01, ** p<0.05, * p<0.1

As reported in Table 4, the equally-weighted WML portfolio yields an average negative return, leading to negative alphas. However, the significant alphas and the highest adjusted R-squared of 0.59, demonstrate that the portfolio return cannot be fully explained by the three factors. Furthermore, we can observe that the estimated coefficients of the CMKT and the CSMB are varied and insignificant in some tests. The CMOM factor has significantly higher coefficients, indicating that the VW-WML portfolios are more exposed to the momentum factors.

Moreover, in Table 7, we adjust the WML portfolios by considering 0.1% static transaction cost (Liu et al., 2022).¹³ Except for the CW-WML portfolio, the estimated alphas for the VW-WML, EQ-WML, and VOL-WML portfolios still significantly deviate from zero. After accounting for transaction costs, portfolio returns cannot be entirely explained by all the three factors considered. The CMOM factor has significantly higher coefficients compared to other factors, confirming the risk exposure to momentum factors. Furthermore, after controlling for standard cryptocurrency factors, the estimated alphas for the VW-WML portfolios are greater than those of other WML portfolios.

4.3. Volume-weighted WML portfolio profitability

Previous findings demonstrate that our VW-WML portfolio return outperforms the cryptocurrency market and all the portfolios applying other common weighting schemes and cannot be explained by the three-factor model. We further test whether the profitability of the VW-WML portfolio originated from the cryptocurrency market, cross-sectional momentum (Liu et al., 2022), or MAX momentum (Li et al., 2021). As introduced in Section 2, we use the volume-weighted market return (CMKTV) and the capital-weighted market return (CMKT). For cross-sectional

¹³To be conservative, we also present the results WML portfolio performance considering doubled static transaction cost of 0.2% in Appendix A. The results are qualitatively the same as the performance when considering 10 bps transaction cost.

momentum, we apply CMOM as in Section 4.2. In addition, we create volume-weighted cross-sectional momentum (CMOMV), which first sorts cryptocurrencies into two portfolios by coin’s trading volume, but not size, compared to CMOM. Following Li et al. (2021), we create MAX momentum portfolios by sorting coins into deciles based on their maximum return in the formation period. We apply both capital and volume-weighted within each group. By buying the top portfolio and selling the bottom portfolio, we can create capital-weighted (CMAXC) and volume-weighted MAX momentum portfolios (CMAXV). All the portfolios are dollar-neutral, hence, we can directly compare them. We examine the VW-WML portfolio profitability by the following equations:

$$R_i = \alpha_i + \beta_i CMKT + \epsilon_i \quad (8)$$

$$R_i = \alpha_i + \beta_i CMKTV + \epsilon_i \quad (9)$$

$$R_i = \alpha_i + \beta_i CMOM + \epsilon_i \quad (10)$$

$$R_i = \alpha_i + \beta_i CMOMV + \epsilon_i \quad (11)$$

$$R_i = \alpha_i + \beta_i CMAXC + \epsilon_i \quad (12)$$

$$R_i = \alpha_i + \beta_i CMAXV + \epsilon_i \quad (13)$$

Table 8 documents the regression results. We find that the VW-WML portfolio profits are derived from but not entirely subsumed by cross-sectional momentum and MAX momentum returns, with significantly positive alphas. The VW-WML returns are more related to CMOMV compared to other momentum factors, with the highest estimated coefficients of 0.76 (1-day formation period), 0.59 (7-day formation period), and 0.69 (14-day formation period). However, the alphas remain positive and significant, therefore, our findings cannot be explained by other forms of momentum.

Table 8: Volume-weighted WML portfolio profitability

This table examines whether the VW-WML portfolio profitability can be explained by the capital-weighted market return (CMKT), volume-weighted market return (CMKTV), cross-section momentum portfolio weighted by market capitalisation (CMOM) and trading volume (CMOMV) following (Liu et al., 2022), and max momentum portfolio weighted by market capitalisation (CMAXC) and trading volume (CMAXV) following (Li et al., 2021). The t-statistics are reported in parentheses.

Panel A: VW-WML portfolios												
	1-day formation period			7-day formation period			14-day formation period					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.009*** (7.12)	0.009*** (5.86)	0.021*** (10.45)	0.021*** (10.66)	0.010*** (7.10)	0.005*** (4.72)	0.009*** (7.45)	0.008*** (6.71)	0.017*** (10.73)	0.016*** (10.53)	0.007*** (6.87)	0.014*** (8.62)
CMKT	0.03 (0.53)						0.05 (1.47)					
CMKTV		0.09 (0.73)						0.13** (2.27)				
CMOM			0.76*** (9.36)						0.61*** (9.91)			
CMOMV				0.76*** (9.70)						0.59*** (9.59)		
CMAXC					0.06* (1.83)						0.01* (1.83)	
CMAXV						0.24*** (12.29)						
Adj. R^2	0.00	0.00	0.10	0.10	0.03	0.27	0.00	0.01	0.10	0.09	0.00	0.10
Panel B: VW-WML portfolios considering transaction cost												
	1-day formation period			7-day formation period			14-day formation period					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.007*** (5.59)	0.007*** (4.53)	0.019*** (9.48)	0.019*** (9.66)	0.008*** (5.66)	0.003*** (2.88)	0.007*** (5.72)	0.006*** (5.01)	0.015*** (9.42)	0.014*** (9.24)	0.005*** (4.99)	0.012*** (7.37)
CMKT	0.03 (0.53)						0.05 (1.47)					
CMKTV		0.09 (0.73)						0.13** (2.27)				
CMOM			0.76*** (9.36)						0.61*** (9.91)			
CMOMV				0.76*** (9.70)						0.59*** (9.59)		
CMAXC					0.06* (1.83)						0.01* (1.83)	
CMAXV						0.24*** (12.29)						
Adj. R^2	0.00	0.00	0.10	0.10	0.03	0.27	0.00	0.01	0.10	0.09	0.00	0.10

*** p<0.01, ** p<0.05, * p<0.1

5. Robustness checks

We conduct several robustness checks with the previous analyses. First, to be more conservative, we conduct robustness checks considering the doubled static transaction cost of 0.2%. Appendix A presents the portfolio returns, Sharpe ratios, and WML portfolios controlled by common cryptocurrency factors compared to CW-WML, EQ-WML, and VOL-WML portfolios. The results are qualitatively the same as the performance when considering 10 bps transaction cost. VW-WML portfolios yield the highest Sharpe ratios of 1.24 (1-day formation period), 1.43 (1-day formation period), and 0.73 (14-day formation period). Further, VW-WML portfolios cannot be subsumed by cryptocurrency common risk factors.

Second, although short-selling has become more available in recent years, Appendix B provides the regression outcomes of the Winner portfolios controlled for factors to evaluate performance excluding short-selling. Similar to our findings in Section 4.2, the estimated alphas remain positive and significant for VW Winner and CW Winner portfolios. However, we can observe that compared to WML portfolios, the CMKT has more explanatory power in Winner portfolios.

Third, we test the WML portfolio performance on sub-sample periods split by the year 2018, when the cryptocurrency becomes more diverse. We discuss the TSMOM of the VW market return in Section 3. Appendix C reports the WML portfolio performance and outcomes after controlling for three risk factors. We find that although the VW-WML average return after 2018 is lower than that of before 2018, the Sharpe ratio is higher after 2018, indicating that the strategy can generate higher adjusted returns after 2018. Additionally, the alphas of VW-WML portfolios remain significantly positive after controlling common cryptocurrency risk factors, showing that the portfolio profits cannot be explained by these factors in sub-samples.

Fourth, to deal with concerns of wash trading in the cryptocurrency market, we use the vol-

ume reported by trusted exchanges from Coin Metrics data. To guarantee the data quality, the selection of trusted exchanges is evaluated by publicly available information, for instance, financial statements and disclosures.¹⁴ The analyses with trusted volume data are presented in Appendix D. The results are qualitatively similar to our main findings.

Fifth, because we have 3,192 coins in our research sample, some coins may not have sufficient liquidity of all time. To be more realistic, we re-examine the strategy with the top 148 coins, which have \$1 million market capitalization every day from 2014 to 2023 and with \$1 million average daily trading volume. To ensure the liquidity of these 148 coins, we calculate the Amihud illiquidity ratio (Amihud, 2002). The median of the daily average illiquidity ratio for these 148 coins is 0.000000005, compared to the figure for the full research sample is 0.000007, ensuring the selected 148 coins are comparably liquid in our research sample. The results are reported in Appendix E. The mean returns and the Sharpe ratios of the VW-WML portfolio are slightly lower than our main findings but still outperform other portfolios. After controlling for three factors, the alphas remain significantly positive and larger than portfolios with other weighting schemes.

Sixth, Bitcoin dominates the market for over 50% of volume weighting most of the time as reported in Section 2. Hence, we re-examine the strategy by excluding Bitcoin. The results are reported in Appendix F. We find higher average VW-WML portfolio returns and Sharpe ratios when excluding Bitcoin. Furthermore, the estimated alphas after controlling for three risk factors are 0.027 (1-day formation period), 0.021 (7-day formation period), and 0.018 (14-day formation period), which is similar to the values of our findings in Section 4.2. All the robustness checks indicate a strong TSMOM effect in the cryptocurrency market and the significant profitability of the VW-WML portfolio.

¹⁴For more information, see <https://coinmetrics.io/special-insights/trusted-exchange-framework-2-1/>

6. Conclusion

This paper contributes to the literature on time series momentum (TSMOM) in the cryptocurrency market. We address the gap between trading volume and TSMOM. Our study documents strong evidence of TSMOM in the cryptocurrency market, in which the effect persists among 3-day to 2-week time horizons. Furthermore, by modifying the TSMOM strategy with trading volume, we construct VW-WML portfolios based on the TSMOM strategy and compare them to WML portfolios weighted by market capitalization, equal proportion, and inverse volatility. The VW-WML dominates all the other WML portfolios and the market portfolio in terms of portfolio cumulative returns. Our portfolio yields the annualised Sharpe ratio of 1.91 (1-day formation period), 2.78 (7-day formation period), and 1.99 (14-day formation period) despite the cryptocurrency market is highly volatile. Considering the transaction costs, the VW-WML portfolio still dominates all the other portfolios considered in our analysis.

Moreover, we find that the standard three-factor cannot well explain the VW-WML portfolio returns, which is inconsistent with findings in Liu et al. (2022). With assessments on the VW-WML portfolio profitability, we observe that the VW-WML portfolio return is associated with but not entirely subsumed by cross-sectional momentum and MAX momentum. Our results indicate that the WML portfolio returns are more risk-exposed to the momentum factor. To deal with the concerns of wash trading and illiquidity, we also conduct analysis by volume reported from trusted exchanges and the top 148 liquid coins for robustness checks.

Our findings can support investors in the cryptocurrency market from several perspectives. First, the TSMOM strategy modified by volume can be profitable and help traders make trading decisions. Furthermore, with the growing number of institutional investors, our study reveals that the volume-weighted TSMOM strategy has more risk exposure to the momentum factor. The VW-

WML portfolio returns are not subsumed by cross-sectional momentum and MAX momentum portfolio profits. The evidence can help institutional investors on managing portfolio risks and rebalancing. Lastly, our study can guide regulators in reassessing cryptocurrency market stability from a trading volume perspective.

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Online Appendix for Cryptocurrency Volume-Weighted Time Series Momentum

A. Winner Minus Loser (WML) portfolios with transaction costs of 0.2% after controlling for cryptocurrency factors

Table .9: Time series momentum strategy Winner Minus Loser (WML) long-short portfolio

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. The WML strategy longs the coins with positive past returns and shorts the coins with negative past returns. Within the long and short portfolios, we weight coins by trading volume, market capitalisation, equal proportion, and inverse volatility. We show the results of lookback periods of 1-day, 7-day, and 14-day. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.2% (Liu and Tsyvinski, 2021). The Sharpe ratios are annualised by 365 days.

Panel A: WML portfolio performance									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.19*** (9.54)	0.25** (2.29)	0.94*** (6.91)	1.09*** (10.20)	0.21** (2.44)	0.87*** (8.45)	1.04*** (9.30)	0.37*** (4.06)	0.67*** (5.94)
Sharpe ratio			2.17			2.66			1.86
Capital-weighted average return (daily, percentage)	0.48*** (4.15)	0.15* (1.79)	0.33*** (2.87)	0.34*** (3.46)	0.06 (0.70)	0.28*** (2.80)	0.37*** (3.83)	0.13* (1.76)	0.24*** (2.65)
Sharpe ratio			0.89			0.87			0.82
Equally-weighted average return (daily, percentage)	-0.76*** (-9.82)	0.43*** (6.00)	-1.19*** (-21.87)	-0.70*** (-9.12)	0.20*** (2.93)	-0.91*** (-16.60)	-0.62*** (-8.05)	0.18** (2.57)	-0.80*** (-14.53)
Sharpe ratio			-6.94			-5.27			-4.62
Volatility-weighted average return (daily, percentage)	- -	- -	- -	-0.12* (-1.86)	-0.06 (-0.87)	-0.07 (-1.13)	-0.17** (-2.38)	-0.01 (-0.18)	-0.15** (-2.53)
Sharpe ratio			-			-0.38			-0.82
Panel B: WML portfolio performance considering transaction cost of 20 bps									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	0.99*** (7.93)	0.05 (0.45)	0.54*** (3.96)	0.89*** (8.32)	0.01 (0.15)	0.47*** (4.58)	0.84*** (7.50)	0.17* (1.88)	0.27** (2.37)
Sharpe ratio			1.24			1.43			0.73
Capital-weighted average return (daily, percentage)	0.28** (2.43)	-0.05 (-0.57)	-0.07 (-0.60)	0.14 (1.42)	-0.14* (-1.69)	-0.12 (-1.20)	0.17* (1.76)	-0.07 (-0.94)	-0.16* (-1.78)
Sharpe ratio			-0.20			-0.39			-0.58
Equally-weighted average return (daily, percentage)	-0.96*** (-12.39)	0.23*** (3.19)	-1.59*** (-29.24)	-0.90*** (-11.70)	0 (0.03)	-1.31*** (-23.92)	-0.82*** (-10.64)	-0.02 (-0.35)	-1.20*** (-21.81)
Sharpe ratio			-9.27			-7.59			-6.92
Volatility-weighted average return (daily, percentage)	- -	- -	- -	-0.32*** (-4.93)	-0.26*** (-4.01)	-0.47*** (-8.02)	-0.37*** (-5.25)	-0.21*** (-3.34)	-0.55*** (-9.12)
Sharpe ratio			-			-2.56			-2.91

*** p<0.01, ** p<0.05, * p<0.1

Table .10: Winner Minus Loser (WML) portfolios with transaction costs (0.2%) after controlling for cryptocurrency factors

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.2% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.02)	0.005*** (4.13)	0.017*** (8.43)	0.018*** (8.55)	-0.001 (-0.95)	-0.001 (-0.58)	0.007*** (4.70)	0.007*** (5.00)	-0.016*** (-24.65)	-0.016*** (-24.96)	-0.004*** (-7.83)	-0.004*** (-7.64)	-	-	-	-
CMKT-Rf	0.03 (0.53)	0.02 (0.31)	0.05 (1.04)	0.03 (0.62)	0.14* (1.88)	0.12* (1.68)	0.15** (2.13)	0.13* (1.85)	0.00 (-0.16)	0.00 (0.12)	0.02* (1.74)	0.02 (1.45)				
CSMB		-0.08* (-1.70)		-0.13*** (-3.06)		-0.12*** (-2.68)		-0.15*** (-3.46)		0.03 (1.47)		-0.02 (-1.49)				
CMOM			0.76*** (9.39)	0.77*** (9.53)			0.48*** (8.62)	0.50*** (9.03)			0.73*** (33.52)	0.74*** (33.87)				
Adj. R^2	0.00	0.00	0.10	0.10	0.01	0.01	0.06	0.07	0.00	0.00	0.59	0.59				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (3.96)	0.005*** (4.01)	0.012*** (8.09)	0.013*** (8.12)	-0.001 (-1.24)	-0.001 (-1.02)	0.004*** (3.66)	0.005*** (3.66)	-0.013*** (-20.69)	-0.013*** (-20.71)	-0.004*** (-7.86)	-0.004*** (-7.81)	-0.004*** (-6.79)	-0.004*** (-6.66)	0.001* (1.86)	0.001* (1.82)
CMKT-Rf	0.05 (1.48)	0.04 (1.24)	0.08*** (2.77)	0.07** (2.37)	0.02 (0.34)	0.01 (0.17)	0.04 (0.88)	0.03 (0.71)	-0.04* (-1.81)	-0.03 (-1.44)	0.00 (-0.08)	0.00 (0.18)	-0.21*** (-5.30)	-0.21*** (-5.08)	-0.19*** (-5.70)	-0.19*** (-5.53)
CSMB		-0.04 (-1.09)		-0.06* (-1.71)		-0.05 (-1.06)		-0.07 (-1.37)		0.05** (2.04)		0.02 (1.47)		0.01 (0.34)		0.00 (-0.19)
CMOM			0.61*** (10.08)	0.62*** (10.09)			0.45*** (8.62)	0.45*** (8.66)			0.73*** (34.73)	0.72*** (34.57)			0.44*** (16.37)	0.44*** (16.27)
Adj. R^2	0.00	0.00	0.10	0.10	0.00	0.00	0.06	0.06	0.00	0.01	0.51	0.51	0.06	0.06	0.23	0.23
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.002** (1.98)	0.003** (2.20)	0.009*** (5.99)	0.010*** (6.20)	-0.002** (-2.14)	-0.002** (-1.68)	0.003*** (2.73)	0.003*** (3.06)	-0.012*** (-18.27)	-0.012*** (-18.10)	-0.004*** (-6.55)	-0.004*** (-6.53)	-0.005*** (-7.94)	-0.005*** (-7.69)	0.000 (0.33)	0.000 (0.49)
CMKT-Rf	0.09*** (2.60)	0.08** (2.18)	0.12*** (3.28)	0.10*** (2.80)	0.11*** (2.70)	0.09*** (2.36)	0.12*** (3.12)	0.10*** (2.80)	-0.02 (-0.80)	-0.02 (-0.84)	0.01 (0.74)	0.01 (0.58)	-0.17*** (-4.29)	-0.18*** (-4.32)	-0.16*** (-4.24)	-0.16*** (-4.29)
CSMB		-0.09** (-2.33)		-0.09*** (-2.82)		-0.12*** (-3.09)		-0.12*** (-3.30)		-0.01 (-0.41)		-0.01 (-1.06)		-0.03 (-1.64)		-0.04*** (-2.21)
CMOM			0.66*** (9.27)	0.66*** (9.32)			0.45*** (8.71)	0.45*** (8.79)			0.76*** (27.14)	0.76*** (27.21)			0.51*** (18.43)	0.51*** (18.52)
Adj. R^2	0.00	0.01	0.10	0.10	0.01	0.01	0.08	0.08	0.00	0.00	0.53	0.53	0.04	0.04	0.24	0.24
*** p<0.01, ** p<0.05, * p<0.1																

*** p<0.01, ** p<0.05, * p<0.1

B. Winner portfolios after controlling for factors

Table .11: Winner portfolios after controlling for factors

The table shows the regression results of the Winner portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted Winner portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period															
VW Winner				CW Winner				EQ Winner				VOL Winner			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.010*** (8.65)	0.009*** (8.33)	0.018*** (9.52)	0.018*** (9.24)	0.003*** (3.01)	0.003*** (2.79)	0.008*** (5.46)	0.008*** (5.72)	-0.009*** (-13.76)	-0.010*** (-17.38)	-0.002*** (-2.58)	-0.003*** (-4.88)	-	-	-
CMKT-Rf	1.00*** (17.60)	1.02*** (20.55)	1.02*** (17.84)	1.03*** (20.36)	1.02*** (13.39)	1.02*** (14.37)	1.03*** (13.55)	1.03*** (14.39)	0.83*** (20.15)	0.88*** (30.59)	0.85*** (20.08)	0.89*** (29.80)	0.89*** (29.80)	0.89*** (29.80)	0.89*** (29.80)
CSMB	0.12** (2.26)			0.08 (1.57)		0.01 (0.21)	-0.01 (-0.25)	-0.01 (-0.25)		0.35*** (5.90)		0.32*** (5.53)			
CMOM			0.53*** (7.00)	0.52*** (6.89)			0.31*** (5.86)	0.31*** (5.85)			0.47*** (18.64)	0.44*** (18.38)			
Adj. R^2	0.29	0.30	0.35	0.35	0.35	0.35	0.37	0.37	0.52	0.60	0.63	0.70			
7-day formation period															
VW Winner				CW Winner				EQ Winner				VOL Winner			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.009*** (9.06)	0.008*** (8.54)	0.013*** (9.67)	0.013*** (9.34)	0.002* (1.87)	0.001 (1.48)	0.005*** (4.56)	0.004*** (4.11)	-0.009*** (-12.12)	-0.010*** (-14.90)	-0.003*** (-4.01)	-0.004*** (-6.24)	-0.002*** (-3.44)	-0.003*** (-4.84)	0.000 (0.05)
CMKT-Rf	0.95*** (18.71)	0.98*** (22.79)	0.98*** (19.08)	1.00*** (23.19)	0.89*** (17.34)	0.90*** (20.40)	0.91*** (17.38)	0.92*** (20.40)	0.79*** (18.47)	0.84*** (27.55)	0.82*** (19.07)	0.87*** (28.99)	0.51*** (11.29)	0.56*** (13.43)	0.57*** (14.71)
CSMB	0.15*** (3.51)			0.14*** (3.24)		0.08 (1.58)	0.07 (1.41)	0.07 (1.41)		0.34*** (7.33)		0.32*** (6.92)		0.26*** (7.35)	0.25*** (7.13)
CMOM			0.37*** (6.32)	0.36*** (6.33)			0.25*** (5.02)	0.25*** (5.01)			0.45*** (15.11)	0.43*** (16.18)		0.26*** (8.44)	0.25*** (8.47)
Adj. R^2	0.37	0.37	0.40	0.41	0.37	0.37	0.39	0.40	0.47	0.55	0.57	0.64	0.28	0.35	0.39
14-day formation period															
VW Winner				CW Winner				EQ Winner				VOL Winner			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.008*** (8.16)	0.008*** (7.83)	0.013*** (8.67)	0.013*** (8.47)	0.002** (2.52)	0.002** (2.33)	0.005*** (4.52)	0.004*** (4.37)	-0.008*** (-11.00)	-0.009*** (-13.03)	-0.003*** (-3.86)	-0.004*** (-5.31)	-0.003*** (-4.13)	-0.004*** (-5.35)	0.000 (-0.55)
CMKT-Rf	0.98*** (19.15)	0.99*** (22.26)	1.00*** (19.01)	1.02*** (22.04)	0.92*** (20.18)	0.93*** (22.54)	0.94*** (20.19)	0.94*** (22.55)	0.80*** (19.63)	0.85*** (27.88)	0.82*** (19.74)	0.87*** (28.53)	0.56*** (12.27)	0.59*** (14.26)	0.61*** (14.78)
CSMB	0.10** (2.07)			0.10** (2.05)		0.02 (0.53)	0.02 (0.48)	0.02 (0.48)		0.28*** (5.41)		0.28*** (5.47)		0.22*** (5.19)	0.22*** (5.26)
CMOM			0.43*** (5.95)	0.42*** (5.92)			0.26*** (5.10)	0.26*** (5.09)			0.46*** (14.07)	0.46*** (14.19)		0.29*** (9.81)	0.29*** (9.87)
Adj. R^2	0.35	0.35	0.39	0.40	0.42	0.42	0.44	0.44	0.49	0.54	0.59	0.65	0.29	0.34	0.39

*** p<0.01, ** p<0.05, * p<0.1

*** p<0.01, ** p<0.05, * p<0.1

C. Sub-sample periods analysis

C.1. Winner Minus Loser (WML) long-short portfolios performance - sub-samples

Table .12: WML long-short portfolios for sampling period from 2014 to 2017

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1%. The Sharpe ratio is annualised by 365 days.

Panel A: WML portfolio performance									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.77*** (9.54)	0.16** (2.29)	1.60*** (6.91)	1.21*** (10.20)	0.14** (2.44)	1.07*** (8.45)	1.18*** (9.30)	0.42*** (4.06)	0.75*** (5.94)
Sharpe ratio			3.07			2.55			1.68
Capital-weighted average return (daily, percentage)	0.63*** (4.15)	0.10* (1.79)	0.54*** (2.87)	0.49*** (3.46)	-0.09 (0.70)	0.57*** (2.80)	0.54*** (3.83)	0.11* (1.76)	0.43*** (2.65)
Sharpe ratio			1.48			1.75			1.32
Equally-weighted average return (daily, percentage)	-0.25*** (-9.82)	0.39*** (6.00)	-0.64*** (-21.87)	-0.19*** (-9.12)	0.16*** (2.93)	-0.35*** (-16.60)	-0.09*** (-8.05)	0.21** (2.57)	-0.31*** (-14.53)
Sharpe ratio			-2.63			-1.44			-1.26
Volatility-weighted average return (daily, percentage)	-	-	-	0.11* (-1.86)	0.03 (-0.87)	0.09 (-1.13)	0.09** (-2.38)	0.13 (-0.18)	-0.03** (-2.53)
Sharpe ratio			-			0.41			-0.17
Panel B: WML portfolio performance considering transaction cost of 10 bps									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.67*** (8.74)	0.06 (1.37)	1.40*** (5.44)	1.11*** (9.26)	0.04 (1.30)	0.87*** (6.52)	1.08*** (8.40)	0.32*** (2.97)	0.55*** (4.15)
Sharpe ratio			2.68			2.07			1.23
Capital-weighted average return (daily, percentage)	0.53*** (3.29)	0 (0.61)	0.34 (1.13)	0.39** (2.44)	-0.19 (-0.50)	0.37 (0.80)	0.44*** (2.79)	0.01 (0.41)	0.23 (0.43)
Sharpe ratio			0.93			1.14			0.7
Equally-weighted average return (daily, percentage)	-0.35*** (-11.10)	0.29*** (4.59)	-0.84*** (-25.55)	-0.29*** (-10.41)	0.06 (1.48)	-0.55*** (-20.26)	-0.19*** (-9.34)	0.11 (1.11)	-0.51*** (-18.17)
Sharpe ratio			-3.45			-2.26			-2.08
Volatility-weighted average return (daily, percentage)	-	-	-	0.01*** (-3.39)	-0.07** (-2.44)	-0.11*** (-4.58)	-0.01*** (-3.81)	0.03* (-1.76)	-0.23*** (-5.82)
Sharpe ratio			-			-0.54			-1.12

*** p<0.01, ** p<0.05, * p<0.1

Table .13: WML long-short portfolios for sampling period from 2018 to 2023

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1%. The Sharpe ratio is annualised by 365 days.

Panel A: WML portfolio performance									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	0.80*** (9.54)	0.31** (2.29)	0.49*** (6.91) 1.39	1.00*** (10.20)	0.26** (2.44)	0.74*** (8.45) 3.00	0.94*** (9.30)	0.34*** (4.06)	0.61*** (5.94) 2.18
Sharpe ratio									
Capital-weighted average return (daily, percentage)	0.38*** (4.15)	0.19* (1.79)	0.19*** (2.87) 0.51	0.24*** (3.46)	0.16 (0.70)	0.08*** (2.80) 0.25	0.26*** (3.83)	0.14* (1.76)	0.11*** (2.65) 0.41
Sharpe ratio									
Equally-weighted average return (daily, percentage)	-1.10*** (-9.82)	0.45*** (6.00)	-1.55*** (-21.87) -16.92	-1.05*** (-9.12)	0.23*** (2.93)	-1.28*** (-16.60) -13.36	-0.98*** (-8.05)	0.15** (2.57)	-1.13*** (-14.53) -11.44
Sharpe ratio									
Volatility-weighted average return (daily, percentage)	-	-	-	-0.28* (-1.86)	-0.11 (-0.87)	-0.17 (-1.13) -1.08	-0.34** (-2.38)	-0.1 (-0.18)	-0.23** (-2.53) -1.34
Sharpe ratio									
Panel B: WML portfolio performance considering transaction cost of 10 bps									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	0.70*** (8.74)	0.21 (1.37)	0.29*** (5.44) 0.82	0.90*** (9.26)	0.16 (1.30)	0.54*** (6.52) 2.18	0.84*** (8.40)	0.24*** (2.97)	0.41*** (4.15) 1.45
Sharpe ratio									
Capital-weighted average return (daily, percentage)	0.28*** (3.29)	0.09 (0.61)	-0.01 (1.13) -0.04	0.14** (2.44)	0.06 (-0.50)	-0.12 (0.80) -0.4	0.16*** (2.79)	0.04 (0.41)	-0.09 (0.43) -0.36
Sharpe ratio									
Equally-weighted average return (daily, percentage)	-1.20*** (-11.10)	0.35*** (4.59)	-1.75*** (-25.55) -19.09	-1.15*** (-10.41)	0.13 (1.48)	-1.48*** (-20.26) -15.44	-1.08*** (-9.34)	0.05 (1.11)	-1.33*** (-18.17) -13.46
Sharpe ratio									
Volatility-weighted average return (daily, percentage)	-	-	-	-0.38*** (-3.39)	-0.21** (-2.44)	-0.37*** (-4.58) -2.31	-0.44*** (-3.81)	-0.20* (-1.76)	-0.43*** (-5.82) -2.46
Sharpe ratio									

*** p<0.01, ** p<0.05, * p<0.1

C.2. WML portfolios after controlling for factors - sub-samples

Table .14: WML portfolios after controlling by factors for sampling period from 2014 to 2017

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.015*** (5.86)	0.016*** (6.05)	0.023*** (8.00)	0.024*** (8.22)	0.005*** (2.81)	0.005*** (3.15)	0.009*** (5.17)	0.010*** (5.56)	-0.007*** (-5.05)	0.000 (0.36)	0.001 (0.63)	-	-	-	-	-
CMKT-Rf	0.24** (2.52)	0.23** (2.51)	0.23** (2.50)	0.22** (2.47)	0.23*** (2.61)	0.23*** (2.60)	0.22*** (2.63)	0.22*** (2.60)	0.05 (1.17)	0.04 (1.18)	0.04 (1.17)	0.04** (-0.04**)	0.02 (0.62)	0.72*** (28.49)	0.72*** (28.79)	0.57
CSMB	-0.08* (-1.88)			-0.14*** (-3.40)		-0.11*** (-3.65)		-0.14*** (-4.73)								
CMOM			0.78*** (8.51)	0.79*** (8.70)			0.47*** (8.50)	0.48*** (8.88)								
Adj. R^2	0.01	0.01	0.15	0.16	0.02	0.02	0.12	0.14	0.00	0.00	0.57	0.57				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.010*** (4.36)	0.011*** (4.42)	0.015*** (6.22)	0.016*** (6.28)	0.005*** (3.10)	0.006*** (3.22)	0.009*** (4.91)	0.009*** (5.02)	-0.003*** (-2.81)	0.002* (1.88)	0.002* (1.68)	0.001 (0.83)	0.001 (0.64)	0.004*** (4.21)	0.004*** (3.99)	0.004*** (-0.02)
CMKT-Rf	0.13* (1.68)	0.13* (1.67)	0.13* (1.92)	0.13* (1.90)	0.10 (1.49)	0.09 (1.48)	0.10* (1.68)	0.10* (1.67)	-0.03 (-0.59)	-0.03 (-0.57)	-0.03 (-0.85)	-0.02 (-0.39)	-0.02 (-0.37)	-0.02 (-0.48)	-0.02 (-0.47)	-0.02 (0.01)
CSMB	-0.06 (-1.25)			-0.08* (-1.92)		-0.05 (-1.40)		-0.07* (-1.91)		0.04 (1.64)	0.02 (0.87)	0.03 (1.24)			0.01 (0.68)	0.01 (0.44***)
CMOM			0.65*** (9.11)	0.66*** (9.15)			0.46*** (7.89)	0.46*** (7.95)			0.71*** (28.10)	0.71*** (27.98)		0.44*** (15.25)	0.44*** (15.18)	0.44*** (0.25)
Adj. R^2	0.00	0.00	0.14	0.14	0.00	0.00	0.11	0.12	0.00	0.00	0.48	0.48	0.00	0.00	0.25	0.25
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.007*** (2.92)	0.008*** (3.14)	0.011*** (4.41)	0.012*** (4.63)	0.004** (2.32)	0.004*** (2.62)	0.006*** (3.68)	0.007*** (3.98)	-0.003** (-2.38)	0.001 (0.96)	0.001 (1.14)	0.000 (-0.36)	0.000 (-0.21)	0.002** (2.43)	0.002** (2.63)	0.003*** (0.02)
CMKT-Rf	0.11 (1.55)	0.11 (1.54)	0.12* (1.73)	0.11* (1.71)	0.11* (1.88)	0.11* (1.87)	0.12** (2.02)	0.11** (2.00)	0.01 (0.24)	0.01 (0.22)	0.01 (0.34)	0.02 (0.45)	0.02 (0.43)	0.02 (0.55)	0.02 (0.53)	0.02 (-0.03**)
CSMB	-0.09** (-2.06)			-0.09** (-2.52)		-0.09*** (-3.05)		-0.09*** (-3.53)		-0.02 (-1.09)	-0.03** (-2.08)	-0.03 (-1.37)			-0.03** (-2.10)	-0.03** (-0.52***)
CMOM			0.75*** (9.28)	0.75*** (9.32)			0.47*** (8.11)	0.47*** (8.18)			0.75*** (22.84)	0.75*** (22.93)		0.52*** (15.75)	0.52*** (15.81)	0.52*** (0.34)
Adj. R^2	0.00	0.00	0.16	0.16	0.00	0.01	0.12	0.13	0.00	0.00	0.52	0.53	0.00	0.00	0.34	0.34

*** p<0.01, ** p<0.05, * p<0.1

*** p<0.01, ** p<0.05, * p<0.1

Table .15: WML portfolios after controlling by factors for sampling period from 2018 to 2023

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (3.99)	0.005*** (3.88)	0.017*** (4.98)	0.017*** (4.95)	0.002 (1.25)	0.002 (1.31)	0.013*** (3.61)	0.013*** (3.67)	-0.015*** (-30.02)	-0.016*** (-30.53)	0.000 (0.98)	0.000 (0.89)	-	-	-	-
CMKT-Rf	-0.08 (-1.23)	-0.13 (-1.33)	-0.04 (-0.77)	-0.09 (-1.06)	0.09 (0.90)	0.07 (0.47)	0.12 (1.33)	0.07 (1.02)	-0.03** (-2.16)	-0.02 (-1.31)	0.02*** (2.68)	0.02*** (3.29)				
CSMB	-0.21 (-1.25)			-0.21 (-1.30)		-0.22 (-1.21)		-0.22 (-1.22)		0.04 (1.59)		0.03** (2.13)				
CMOM			0.58*** (4.18)	0.59*** (4.24)			0.57*** (3.23)	0.57*** (3.28)			0.79*** (29.05)	0.79*** (28.67)				
Adj. R^2	0.00	0.01	0.02	0.03	0.00	0.01	0.02	0.03	0.01	0.01	0.63	0.63				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.007*** (6.87)	0.007*** (6.78)	0.015*** (8.51)	0.015*** (8.59)	0.001 (0.66)	0.001 (0.67)	0.007*** (3.97)	0.007*** (4.08)	-0.013*** (-23.67)	-0.013*** (-23.71)	-0.001 (-0.95)	0.000 (-0.87)	-0.001* (-1.86)	-0.001 (-1.56)	0.005*** (4.01)	0.005*** (4.09)
CMKT-Rf	0.01 (0.25)	0.00 (0.01)	0.05* (1.79)	0.05 (1.43)	-0.03 (-0.45)	-0.05 (-1.21)	0.01 (0.09)	-0.01 (-0.32)	-0.05** (-2.32)	-0.05* (-1.80)	0.02** (2.05)	0.03*** (3.13)	-0.31*** (-6.17)	-0.35*** (-6.25)	-0.27*** (-6.69)	-0.31*** (-6.65)
CSMB		-0.03 (-0.42)		0.00 (0.04)		-0.11 (-0.55)		-0.08 (-0.41)		0.00 (0.14)		0.06*** (3.06)		-0.19*** (-3.04)		-0.16*** (-2.88)
CMOM			0.47*** (5.55)	0.47*** (5.66)			0.38*** (3.36)	0.37*** (3.25)			0.76*** (23.22)	0.77*** (23.33)		0.41*** (6.38)	0.39*** (6.48)	
Adj. R^2	0.00	0.00	0.03	0.03	0.00	0.00	0.01	0.01	0.01	0.01	0.59	0.59	0.18	0.20	0.24	0.25
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (5.00)	0.006*** (5.02)	0.011*** (5.33)	0.011*** (5.36)	0.001 (1.01)	0.001 (1.24)	0.007*** (4.14)	0.006*** (4.11)	-0.011*** (-20.06)	-0.011*** (-20.13)	0.000 (-0.34)	0.000 (-0.33)	-0.002*** (-2.70)	-0.002*** (-2.45)	0.005*** (4.34)	0.005*** (4.35)
CMKT-Rf	0.08** (2.12)	0.06 (1.24)	0.10*** (2.64)	0.08* (1.80)	0.10* (1.94)	0.03 (0.76)	0.12** (2.34)	0.06 (1.42)	-0.03 (-1.53)	-0.03 (-1.18)	0.01 (1.15)	0.02** (2.03)	-0.27*** (-5.18)	-0.31*** (-5.13)	-0.25*** (-5.07)	-0.28*** (-4.96)
CSMB		-0.12 (-1.38)		-0.10 (-1.27)		-0.31* (-1.88)		-0.29* (-1.79)		0.01 (0.47)		0.05** (2.50)		-0.16** (-2.42)		-0.14** (-2.23)
CMOM			0.34*** (2.95)	0.34*** (2.95)			0.40*** (3.43)	0.38*** (3.32)			0.78*** (20.16)	0.78*** (20.14)		0.47*** (10.17)	0.46*** (10.27)	
Adj. R^2	0.00	0.01	0.02	0.02	0.01	0.02	0.03	0.04	0.00	0.00	0.54	0.54	0.12	0.13	0.17	0.18

*** p<0.01, ** p<0.05, * p<0.1

*** p<0.01, ** p<0.05, * p<0.1

C.3. WML portfolios with transaction costs after controlling for factors - sub-samples

Table .16: WML portfolios with transaction cost after controlling for factors, 2014-2017

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

1-day formation period												
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.013*** (5.10)	0.014*** (5.29)	0.021*** (7.29)	0.022*** (7.52)	0.003 (1.61)	0.003** (1.97)	0.007*** (4.04)	0.008*** (4.46)	-0.009*** (-6.58)	-0.002*** (-1.86)	-0.001 (-1.56)	-
CMKT-Rf	0.24** (2.52)	0.23** (2.51)	0.23** (2.50)	0.22** (2.47)	0.23*** (2.61)	0.23*** (2.60)	0.22*** (2.63)	0.22*** (2.60)	0.05 (1.17)	0.05 (1.18)	0.04 (1.17)	-
CSMB	-0.08* (-1.88)	-0.08* (-1.88)	-0.14*** (-3.40)	-0.14*** (-3.40)	-0.11*** (-3.65)	-0.11*** (-3.65)	-0.14*** (-4.73)	-0.14*** (-4.73)	0.02 (0.62)	0.02 (0.62)	-0.04** (-2.10)	-
CMOM			0.78*** (8.51)	0.79*** (8.70)			0.47*** (8.50)	0.48*** (8.88)			0.72*** (28.49)	0.72*** (28.79)
Adj. R^2	0.01	0.01	0.15	0.16	0.02	0.02	0.12	0.14	0.00	0.00	0.57	0.57
7-day formation period												
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.008*** (3.52)	0.009*** (3.60)	0.013*** (5.40)	0.014*** (5.48)	0.003** (1.96)	0.004** (2.10)	0.007*** (3.79)	0.007*** (3.94)	-0.005*** (-4.44)	-0.006*** (-4.53)	0.000 (-0.28)	0.002** (2.18)
CMKT-Rf	0.13* (1.68)	0.13* (1.67)	0.13* (1.92)	0.13* (1.90)	0.10 (1.49)	0.09 (1.48)	0.10* (1.68)	0.10* (1.67)	-0.03 (-0.59)	-0.03 (-0.57)	-0.02 (-0.39)	-0.02 (-0.47)
CSMB	-0.06 (-1.25)	-0.06 (-1.25)	-0.08* (-1.92)	-0.08* (-1.92)	-0.05 (-1.40)	-0.05 (-1.40)	-0.07* (-1.91)	-0.07* (-1.91)	0.04 (1.64)	0.04 (1.64)	0.03 (1.24)	0.01 (0.68)
CMOM			0.65*** (9.11)	0.66*** (9.15)			0.46*** (7.89)	0.46*** (7.95)			0.71*** (28.10)	0.71*** (27.98)
Adj. R^2	0.00	0.00	0.14	0.14	0.00	0.00	0.11	0.12	0.00	0.00	0.48	0.48
14-day formation period												
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005** (2.11)	0.006** (2.33)	0.009*** (3.63)	0.010*** (3.86)	0.002 (1.14)	0.003 (1.45)	0.004** (2.55)	0.005*** (2.86)	-0.005*** (-3.92)	-0.005*** (-3.77)	-0.002** (-2.16)	0.001 (0.43)
CMKT-Rf	0.11 (1.55)	0.11 (1.54)	0.12* (1.73)	0.11* (1.71)	0.11* (1.88)	0.11* (1.87)	0.12** (2.02)	0.11** (2.00)	0.01 (0.24)	0.01 (0.22)	0.02 (0.45)	0.02 (0.55)
CSMB	-0.09** (-2.06)	-0.09** (-2.06)	-0.09** (-2.52)	-0.09** (-2.52)	-0.09*** (-3.05)	-0.09*** (-3.05)	-0.09*** (-3.53)	-0.09*** (-3.53)	-0.02 (-1.09)	-0.02 (-1.09)	-0.03** (-1.37)	-0.03** (-2.10)
CMOM			0.75*** (9.28)	0.75*** (9.32)			0.47*** (8.11)	0.47*** (8.18)			0.75*** (22.84)	0.75*** (22.93)
Adj. R^2	0.00	0.00	0.16	0.16	0.00	0.01	0.12	0.13	0.00	0.00	0.52	0.53
*** p<0.01, ** p<0.05, * p<0.1												

Table .17: WML portfolios with transaction cost after controlling for factors, 2018-2023

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

1-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.003** (2.39)	0.003** (2.37)	0.015*** (4.38)	0.015*** (4.36)	0.000 (-0.18)	0.000 (-0.05)	0.011*** (3.12)	-0.018*** (-33.88)	-0.018*** (-34.45)	-0.002*** (-3.54)	-0.002*** (-3.55)	-	-	-	-
CMKT-Rf	-0.08 (-1.23)	-0.13 (-1.33)	-0.04 (-0.77)	-0.09 (-1.06)	0.09 (0.90)	0.12 (1.33)	0.07 (1.02)	-0.03** (-2.16)	-0.02 (-1.31)	0.02*** (2.68)	0.02*** (3.29)				
CSMB	-0.21 (-1.25)			-0.21 (-1.30)	-0.22 (-1.21)		-0.22 (-1.22)		0.04 (1.59)		0.03** (2.13)				
CMOM			0.58*** (4.18)	0.59*** (4.24)		0.57*** (3.23)	0.57*** (3.28)			0.79*** (29.05)	0.79*** (28.67)				
Adj. R^2	0.00	0.01	0.02	0.03	0.00	0.01	0.02	0.01	0.01	0.63	0.63				
7-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.99)	0.005*** (4.93)	0.013*** (7.36)	0.013*** (7.44)	-0.001 (-0.97)	-0.001 (-0.83)	0.005*** (2.83)	-0.015*** (-27.38)	-0.015*** (-27.42)	-0.003*** (-5.03)	-0.002*** (-5.04)	-0.003*** (-4.71)	-0.003*** (-4.26)	0.003** (2.49)	0.003** (2.50)
CMKT-Rf	0.01 (0.25)	0.00 (0.01)	0.05* (1.79)	0.05 (1.43)	-0.03 (-0.45)	-0.05 (-1.21)	0.01 (0.09)	-0.05** (-2.32)	-0.05* (-1.80)	0.02** (2.05)	0.03*** (3.13)	-0.31*** (-6.17)	-0.35*** (-6.25)	-0.27*** (-6.69)	-0.31*** (-6.65)
CSMB	-0.03 (-0.42)			0.00 (0.04)	-0.11 (-0.55)		-0.08 (-0.41)		0.00 (0.14)		0.06*** (3.06)	-0.19*** (-3.04)	-0.19*** (-3.04)	-0.16*** (-2.88)	-0.16*** (-2.88)
CMOM			0.47*** (5.55)	0.47*** (5.66)		0.38*** (3.36)	0.37*** (3.25)			0.76*** (23.22)	0.77*** (23.33)			0.41*** (6.38)	0.39*** (6.48)
Adj. R^2	0.00	0.00	0.03	0.03	0.00	0.00	0.01	0.01	0.01	0.59	0.59	0.18	0.20	0.24	0.25
14-day formation period															
VW-WML				CW-WML				EQ-WML				VOL-WML			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.004*** (3.30)	0.004*** (3.33)	0.009*** (4.34)	0.009*** (4.37)	-0.001 (-1.16)	-0.001 (-0.87)	0.005*** (2.88)	-0.013*** (-23.62)	-0.013*** (-23.69)	-0.002*** (-4.10)	-0.002*** (-4.09)	-0.004*** (-5.37)	-0.004*** (-5.03)	0.003** (2.47)	0.003** (2.47)
CMKT-Rf	0.08** (2.12)	0.06 (1.24)	0.10*** (2.64)	0.08* (1.80)	0.10* (1.94)	0.03 (0.76)	0.12** (2.34)	-0.03 (-1.53)	-0.03 (-1.18)	0.01 (1.15)	0.02** (2.03)	-0.27*** (-5.18)	-0.31*** (-5.13)	-0.25*** (-5.07)	-0.28*** (-4.96)
CSMB	-0.12 (-1.38)			-0.10 (-1.27)	-0.31* (-1.88)		-0.29* (-1.79)		0.01 (0.47)		0.05** (2.50)	-0.16** (-2.42)	-0.16** (-2.42)	-0.14** (-2.23)	-0.14** (-2.23)
CMOM			0.34*** (2.95)	0.34*** (2.95)		0.40*** (3.43)	0.38*** (3.32)			0.78*** (20.16)	0.78*** (20.14)			0.47*** (10.17)	0.46*** (10.27)
Adj. R^2	0.00	0.01	0.02	0.02	0.01	0.02	0.03	0.00	0.00	0.54	0.54	0.12	0.13	0.17	0.18
*** p<0.01, ** p<0.05, * p<0.1															

*** p<0.01, ** p<0.05, * p<0.1

D. Coin Metrics data

D.1. Summary statistics

Table .18: Coin Metrics summary statistics

The two tables tabulate the summary statistics of Coin Metrics data with volume reported by trusted exchanges. Panel A provides the number of coins and the mean and median of market capitalisation and daily trading volume in USD by year. We only consider coins with a market capitalisation above \$1 million daily sampling from January 2014 to December 2023. Panel B shows the summary statistics of the daily return of the volume-weighted market return, Bitcoin, Ethereum, and XRP.

Panel A: Sample data					
Year	Count	Market Cap (mil)		Volume (thous)	
		Mean	Median	Mean	Median
2014	8	1,521.51	31.08	3,242.77	15.41
2015	12	588.11	15.95	2,156.78	12.27
2016	16	849.57	57.00	6,052.79	145.15
2017	42	4,746.24	432.81	72,468.55	3,197.63
2018	51	7,322.76	453.20	84,644.94	1,907.08
2019	58	4,058.24	156.45	45,542.90	825.73
2020	74	5,179.53	267.41	74,475.98	2,049.98
2021	75	22,771.08	1,507.10	421,568.46	30,489.60
2022	74	14,057.33	396.65	201,499.09	9,072.67
2023	70	13,903.21	305.14	145,801.65	5,014.94
Full	78	7,497.30	261.13	105,709.53	1,743.36
Panel B: Volume-weighted market return and major cryptocurrencies					
	Mean	Median	STD	Skewness	Kurtosis
VW market	0.49%	0.43%	0.04	-0.11	10.67
CW market	0.15%	0.21%	0.04	-0.82	10.11
EQ market	0.08%	0.31%	0.05	-1.01	10.45
Bitcoin	0.11%	0.12%	0.04	-0.69	10.76
Ethereum	0.25%	0.03%	0.06	-0.18	7.76
XRP	0.14%	-0.09%	0.07	1.57	29.61

D.2. TSMOM regression

Table .19: Time-series momentum of the volume-weighted market return, Coin Metrics

This table presents the TSMOM results for the volume-weighted market return by regressing the current market return on the past cumulative market return among different time horizons. We show the results of cumulative market return from 1-day to 2-week. The t-statistics are shown in brackets.

2014-01-01 - 2023-12-31							
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7
Rt	-0.023	0.010	0.024**	0.028***	0.024***	0.030***	0.024***
	(-1.04)	(0.70)	(2.07)	(2.91)	(2.68)	(3.91)	(3.27)
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.01	0.00
	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.022***	0.021***	0.023***	0.022***	0.020***	0.019***	0.019***
	(3.25)	(3.31)	(3.97)	(3.95)	(3.82)	(3.80)	(3.92)
Adj. R^2	0.00	0.00	0.01	0.01	0.01	0.01	0.01

*** p<0.01, ** p<0.05, * p<0.1

D.3. TSMOM by groups

Table .20: Time-series momentum by groups, Coin Metrics

Following Liu and Tsyvinski (2021), the two tables estimate the time-series momentum magnitude of the volume-weighted market return. Panel A presents the results by grouping volume-weighted market past cumulative returns into terciles labelled as Low, Middle, and High, and evaluating the subsequent day's return. Panel B groups the volume-weighted cumulative returns by the cutoffs calculated by the first two years' data (from 2014 to 2015), and evaluates the out-of-sample (from 2016 to 2023) TSMOM magnitude of the cryptocurrencies market. The results are presented in percentage expression and the t-statistics are reported in brackets. Diff refers to the difference in mean return between the High group and the Low group.

Panel A: Time-series momentum by groups (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-3.45*** (-35.30)	0.32** (2.39)	0.13 (0.98)	0.12 (0.89)	0.16 (1.24)	0.09 (0.65)	-0.03 (-0.24)	0.02 (0.16)	0.04 (0.29)
Middle	0.40*** (24.50)	0.35*** (3.20)	0.44*** (3.98)	0.38*** (3.70)	0.26** (2.44)	0.44*** (4.19)	0.35*** (3.39)	0.28*** (2.67)	0.33*** (3.08)
High	4.43*** (46.99)	0.69*** (5.34)	0.81*** (6.46)	0.87*** (6.70)	0.93*** (7.04)	0.86*** (6.57)	1.06*** (8.02)	1.07*** (8.32)	0.99*** (7.35)
Diff	7.89	0.36	0.67	0.75	0.77	0.77	1.09	1.05	0.96
Weekly HML	19.58*** (34.83)	0.89** (2.04)	1.63*** (3.60)	1.80*** (3.78)	1.84*** (3.75)	1.84*** (3.60)	2.64*** (5.26)	2.51*** (5.08)	2.33*** (4.66)
Panel B: Time-series momentum by groups with the first 2-year cutoffs (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-4.04*** (-32.12)	0.45** (2.58)	0.34* (1.85)	0.17 (0.82)	0.15 (0.74)	0.27 (1.30)	0.18 (0.83)	0.13 (0.57)	0.19 (0.84)
Middle	0.05*** (2.77)	0.57*** (4.45)	0.26* (1.84)	0.47*** (3.63)	0.41*** (2.87)	0.38** (2.45)	0.23* (1.80)	0.11 (0.86)	0.12 (0.97)
High	4.08*** (42.15)	0.69*** (5.26)	0.92*** (7.63)	0.88*** (7.41)	0.89*** (7.68)	0.83*** (7.47)	0.94*** (8.06)	1.02*** (9.00)	0.95*** (8.31)
Diff	8.11	0.23	0.58	0.72	0.75	0.56	0.77	0.89	0.76
Weekly HML	20.60*** (32.18)	1.14** (2.08)	2.31*** (4.23)	2.62*** (4.70)	2.88*** (4.81)	2.61*** (4.15)	3.13*** (5.22)	3.51*** (6.07)	3.34*** (5.66)

*** p<0.01, ** p<0.05, * p<0.1

D.4. WML portfolio performance

Table .21: WML long-short portfolios, Coin Metrics

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1%. The Sharpe ratio is annualised by 365 days.

Panel A: WML portfolio performance									
Lookback			1-day			7-day			14-day
	Winner	Loser	WML	Winner	Loser	WML	Winner	Loser	WML
Volume-weighted average return (daily, percentage)	0.72***	0.16*	0.56***	0.74***	0.10	0.63***	0.66***	0.18**	0.47***
	(6.80)	(1.85)	(4.76)	(7.56)	(1.31)	(5.91)	(7.17)	(2.04)	(4.69)
Sharpe ratio			1.39			1.85			1.34
Capital-weighted average return (daily, percentage)	0.24***	-0.04	0.28***	0.17***	-0.05	0.23***	0.19***	-0.03	0.23***
	(3.43)	(-0.16)	(3.30)	(2.86)	(-0.40)	(3.05)	(3.32)	(-0.15)	(3.23)
Sharpe ratio			0.81			0.83			0.82
Equally-weighted average return (daily, percentage)	-0.06	0.09	-0.15	0.07*	-0.06	0.13**	0.03	-0.06	0.09**
	(0.70)	(1.08)	(-0.26)	(1.85)	(-0.68)	(2.48)	(1.63)	(-0.53)	(2.12)
Sharpe ratio			-0.51			0.52			0.36
Volatility-weighted average return (daily, percentage)	-	-	-	0.10**	0.05	0.05	0.11**	0.05	0.06
	-	-	-	(2.27)	(1.22)	(1.48)	(2.40)	(1.23)	(1.60)
Sharpe ratio			-			0.20			0.25
Panel B: WML portfolio performance considering transaction costs									
Lookback			1-day			7-day			14-day
	Winner	Loser	WML	Winner	Loser	WML	Winner	Loser	WML
Volume-weighted average return (daily, percentage)	0.62***	0.06	0.36***	0.64***	0.00	0.43***	0.56***	0.08	0.27**
	(5.78)	(0.63)	(2.91)	(6.38)	(-0.05)	(3.74)	(5.98)	(0.77)	(2.56)
Sharpe ratio			0.89			1.26			0.77
Capital-weighted average return (daily, percentage)	0.14**	-0.14	0.08	0.07	-0.15*	0.03	0.09*	-0.13*	0.03
	(2.30)	(-1.49)	(1.20)	(1.48)	(-1.89)	(0.45)	(1.96)	(-1.66)	(0.68)
Sharpe ratio			0.22			0.08			0.08
Equally-weighted average return (daily, percentage)	-0.16	-0.01	-0.35**	-0.03	-0.16**	-0.07	-0.07	-0.16**	-0.11
	(-0.48)	(-0.24)	(-2.56)	(0.52)	(-2.10)	(-0.18)	(0.29)	(-1.98)	(-0.56)
Sharpe ratio			-1.17			-0.28			-0.46
Volatility-weighted average return (daily, percentage)	-	-	-	0.00	-0.05	-0.15	0.01	-0.05	-0.14
	-	-	-	(0.95)	(-0.55)	(-1.40)	(1.06)	(-0.54)	(-1.29)
Sharpe ratio			-			-0.66			-0.60

*** p<0.01, ** p<0.05, * p<0.1

D.5. WML portfolios after controlling for factors

Table .22: WML portfolios after controlling for factors, Coin Metrics

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

	1-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.52)	0.006*** (4.77)	0.006*** (5.36)	0.006*** (5.35)	0.003*** (2.44)	0.003*** (2.68)	0.003*** (2.92)	0.003*** (2.97)	-0.002 (-1.53)	-0.001 (-1.32)	-0.001* (-1.81)	-0.001* (-1.71)
CMKT-Rf	0.05 (1.15)	0.06 (1.17)	0.05 (1.29)	0.05 (1.30)	0.07 (1.61)	0.07 (1.60)	0.06* (1.76)	0.06* (1.76)	0.00 (-0.11)	0.00 (0.08)	-0.01 (-0.29)	0.00 (-0.19)
CSMB		0.25* (1.79)		0.04 (0.58)		0.25* (1.75)		0.08 (0.98)	0.35** (2.21)		0.15* (1.74)	
CMOM			0.82*** (10.81)	0.81*** (13.09)			0.67*** (7.55)	0.64*** (10.19)		0.83*** (8.54)	0.78*** (13.85)	
Adj. R^2	0.00	0.03	0.26	0.26	0.00	0.04	0.24	0.25	0.00	0.10	0.47	0.49
	7-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (5.98)	0.006*** (5.93)	0.006*** (6.46)	0.006*** (6.18)	0.002*** (2.63)	0.002*** (2.60)	0.002*** (2.56)	0.002*** (2.40)	0.001* (1.85)	0.001* (1.92)	0.001** (2.04)	0.001** (2.07)
CMKT-Rf	0.00 (0.04)	0.00 (0.05)	-0.01 (-0.15)	-0.01 (-0.15)	0.03 (0.63)	0.03 (0.63)	0.02 (0.57)	0.02 (0.54)	-0.04 (-1.15)	-0.04 (-1.10)	-0.05* (-1.73)	-0.05* (-1.72)
CSMB		0.04 (0.49)		-0.01 (-0.14)		-0.03 (-0.75)		-0.11*** (-2.61)		0.07 (1.56)		0.01 (0.40)
CMOM			0.68*** (10.13)	0.68*** (10.58)			0.43*** (9.10)	0.43*** (9.39)		0.69*** (23.90)	0.69*** (24.39)	
Adj. R^2	0.00	0.00	0.23	0.23	0.00	0.00	0.15	0.16	0.00	0.01	0.45	0.45
	14-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.03)	0.005*** (3.99)	0.004*** (4.10)	0.004*** (4.03)	0.002*** (2.47)	0.002*** (2.44)	0.002*** (2.33)	0.002*** (2.26)	0.001 (1.21)	0.001 (1.21)	0.001 (0.97)	0.001 (0.99)
CMKT-Rf	0.03 (0.59)	0.03 (0.59)	0.02 (0.44)	0.02 (0.43)	0.05 (1.15)	0.05 (1.15)	0.05 (1.09)	0.04 (1.09)	-0.05 (-1.21)	-0.05 (-1.19)	-0.06* (-1.87)	-0.06* (-1.86)
CSMB		-0.03 (-0.56)		-0.04 (-0.76)		-0.03 (-0.62)		-0.04 (-0.83)		0.02 (0.51)		0.02 (0.53)
CMOM			0.65*** (11.90)	0.65*** (11.73)			0.49*** (10.86)	0.49*** (10.67)		0.67*** (18.70)	0.67*** (18.70)	
Adj. R^2	0.00	0.00	0.19	0.19	0.00	0.00	0.18	0.18	0.00	0.00	0.41	0.41
*** p<0.01, ** p<0.05, * p<0.1												

Table .23: WML portfolios with transaction costs after controlling for factors, Coin Metrics

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

	1-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.004*** (2.87)	0.004*** (3.09)	0.004*** (3.47)	0.004*** (3.48)	0.001 (0.60)	0.001 (0.80)	0.001 (0.83)	0.001 (0.89)	-0.004*** (-3.54)	-0.003*** (-3.45)	-0.003*** (-4.48)	-0.003*** (-4.43)
CMKT-Rf	0.05 (1.15)	0.06 (1.17)	0.05 (1.29)	0.05 (1.30)	0.07 (1.61)	0.07 (1.60)	0.06* (1.76)	0.06* (1.76)	0.00 (-0.11)	0.00 (0.08)	-0.01 (-0.29)	0.00 (-0.19)
CSMB		0.25* (1.79)		0.04 (0.58)		0.25* (1.75)		0.08 (0.98)		0.35** (2.21)		0.15* (1.74)
CMOM			0.82*** (10.81)	0.81*** (13.09)			0.67*** (7.55)	0.64*** (10.19)			0.83*** (8.54)	0.78*** (13.85)
Adj. R^2	0.00	0.03	0.26	0.26	0.00	0.04	0.24	0.25	0.00	0.10	0.47	0.49

	7-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.004*** (4.08)	0.004*** (4.06)	0.004*** (4.35)	0.004*** (4.16)	0.000 (0.21)	0.000 (0.18)	0.000 (0.02)	0.000 (-0.18)	-0.001 (-0.86)	-0.001 (-0.78)	-0.001 (-1.47)	-0.001 (-1.42)
CMKT-Rf	0.00 (0.04)	0.00 (0.05)	-0.01 (-0.15)	-0.01 (-0.15)	0.03 (0.63)	0.03 (0.63)	0.02 (0.57)	0.02 (0.54)	-0.04 (-1.15)	-0.04 (-1.10)	-0.05* (-1.73)	-0.05* (-1.72)
CSMB		0.04 (0.49)		-0.01 (-0.14)		-0.03 (-0.75)		-0.11*** (-2.61)		0.07 (1.56)		0.01 (0.40)
CMOM			0.68*** (10.13)	0.68*** (10.58)			0.43*** (9.10)	0.43*** (9.39)			0.69*** (23.90)	0.69*** (24.39)
Adj. R^2	0.00	0.00	0.23	0.23	0.00	0.00	0.15	0.16	0.00	0.01	0.45	0.45

	14-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.003** (2.29)	0.003** (2.26)	0.002** (2.19)	0.002** (2.13)	0.000 (0.17)	0.000 (0.15)	0.000 (-0.15)	0.000 (-0.22)	-0.001 (-1.33)	-0.001 (-1.29)	-0.001** (-2.13)	-0.001** (-2.07)
CMKT-Rf	0.03 (0.59)	0.03 (0.59)	0.02 (0.44)	0.02 (0.43)	0.05 (1.15)	0.05 (1.15)	0.05 (1.09)	0.04 (1.09)	-0.05 (-1.21)	-0.05 (-1.19)	-0.06* (-1.87)	-0.06* (-1.86)
CSMB		-0.03 (-0.56)		-0.04 (-0.76)		-0.03 (-0.62)		-0.04 (-0.83)		0.02 (0.51)		0.02 (0.53)
CMOM			0.65*** (11.90)	0.65*** (11.73)			0.49*** (10.86)	0.49*** (10.67)			0.67*** (18.70)	0.67*** (18.81)
Adj. R^2	0.00	0.00	0.19	0.19	0.00	0.00	0.18	0.18	0.00	0.00	0.41	0.41

*** p<0.01, ** p<0.05, * p<0.1

D.6. Winner portfolios after controlling for factors

Table .24: Winner portfolios after controlling for factors, Coin Metrics

The table shows the regression results of the Winner portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted Winner portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period												
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (5.49)	0.006*** (6.32)	0.006*** (6.31)	0.006*** (6.84)	0.001 (1.02)	0.001 (1.42)	0.001 (1.27)	0.001 (1.53)	-0.002** (-2.15)	-0.002** (-2.26)	-0.002** (-2.36)	-0.002** (-2.57)
CMKT-Rf	1.01*** (30.68)	1.02*** (28.15)	1.02*** (32.56)	1.02*** (31.66)	0.98*** (31.07)	0.99*** (28.91)	0.99*** (32.71)	0.99*** (31.89)	0.94*** (37.40)	0.95*** (39.65)	0.94*** (33.85)	0.95*** (41.22)
CSMB		0.38*** (3.72)		0.25*** (4.31)		0.33*** (2.82)		0.22*** (2.97)		0.58*** (6.79)		0.47*** (10.38)
CMOM			0.60*** (6.39)	0.52*** (9.27)			0.51*** (4.88)	0.43*** (7.37)			0.58*** (4.99)	0.43*** (10.25)
Adj. R^2	0.29	0.37	0.44	0.47	0.35	0.42	0.49	0.52	0.35	0.60	0.55	0.70
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (7.46)	0.006*** (7.53)	0.006*** (7.92)	0.006*** (8.07)	0.000 (0.57)	0.000 (0.63)	0.000 (0.39)	0.000 (0.50)	-0.001 (-1.02)	0.000 (-0.69)	-0.001 (-1.39)	0.000 (-0.45)
CMKT-Rf	0.94*** (27.20)	0.94*** (27.01)	0.94*** (28.43)	0.94*** (28.69)	0.92*** (28.66)	0.92*** (28.65)	0.92*** (29.80)	0.92*** (29.85)	0.89*** (28.17)	0.89*** (30.21)	0.89*** (27.20)	0.89*** (30.89)
CSMB		0.15* (1.86)		0.19** (2.33)		0.04** (2.22)		0.04* (1.73)		0.28*** (3.44)		0.37*** (17.48)
CMOM			0.44*** (5.53)	0.43*** (6.00)			0.24*** (8.06)	0.24*** (7.79)			0.40*** (12.00)	0.38*** (14.68)
Adj. R^2	0.36	0.37	0.47	0.48	0.50	0.50	0.55	0.55	0.44	0.52	0.57	0.67
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (5.95)	0.005*** (6.09)	0.005*** (6.11)	0.005*** (6.46)	0.001 (0.78)	0.001 (0.86)	0.000 (0.50)	0.000 (0.70)	-0.001 (-1.47)	-0.001 (-1.22)	-0.001** (-1.99)	-0.001 (-1.33)
CMKT-Rf	0.97*** (26.95)	0.97*** (26.85)	0.97*** (27.57)	0.97*** (27.76)	0.92*** (28.68)	0.93*** (28.52)	0.92*** (29.54)	0.93*** (29.50)	0.89*** (26.66)	0.89*** (28.22)	0.88*** (25.80)	0.89*** (28.69)
CSMB		0.12** (2.26)		0.17*** (3.82)		0.06* (1.65)		0.09** (2.20)		0.25*** (3.08)		0.36*** (8.05)
CMOM			0.40*** (8.29)	0.40*** (8.74)			0.29*** (7.29)	0.29*** (7.59)			0.38*** (9.58)	0.38*** (11.19)
Adj. R^2	0.39	0.40	0.47	0.49	0.49	0.49	0.55	0.55	0.44	0.51	0.55	0.64
Adj. R^2					0.47	0.50	0.55	0.55	0.44	0.51	0.55	0.64
Adj. R^2									0.46	0.50	0.46	0.51

*** p<0.01, ** p<0.05, * p<0.1

E. Top 148 coins

E.1. Summary statistics

Table .25: Top 148 coins summary statistics

The two tables tabulate the summary statistics of the top 148 coins, which have \$1 million market capitalisation every day from 2014 to 2023 and with \$1 million average daily trading volume. Panel A provides the number of coins and the mean and median of market capitalisation and daily trading volume in USD by year. Panel B shows the summary statistics of the daily return of the volume-weighted market return, Bitcoin, Ethereum, and XRP.

Panel A: Sample data					
Year	Count	Market Cap (mil)		Volume (thous)	
		Mean	Median	Mean	Median
2014	8	1,252.90	41.04	5,865.29	480.25
2015	9	521.95	16.51	4,569.78	139.99
2016	12	976.82	44.75	11,028.49	597.73
2017	39	3,923.91	196.01	144,291.51	4,489.41
2018	44	5,601.78	129.42	257,423.70	3,857.15
2019	44	4,075.97	55.49	666,601.67	2,111.78
2020	69	5,400.24	58.06	1,231,750.34	5,739.78
2021	115	15,529.51	247.26	1,244,055.52	31,547.00
2022	128	7,823.31	89.03	472,927.15	13,546.86
2023	148	6,813.24	87.13	247,850.87	7,268.06
Full	148	5,190.86	80.09	428,741.99	3,781.00
Panel B: Volume-weighted market return and major cryptocurrencies					
	Mean	Median	STD	Skewness	Kurtosis
VW market	0.41%	0.34%	0.04	-0.62	9.24
CW market	0.14%	0.18%	0.04	-1.01	10.84
EQ market	0.07%	0.25%	0.05	-1.21	10.79
Bitcoin	0.11%	0.11%	0.04	-0.76	10.95
Ethereum	0.26%	0.04%	0.06	0.01	8.74
XRP	0.09%	-0.17%	0.06	1.81	35.18

E.2. TSMOM regression

Table .26: Time-series momentum of the volume-weighted market return, Top 148 coins

This table presents the TSMOM results for the volume-weighted market return by regressing the current market return on the past cumulative market return among different time horizons. We show the results of cumulative market return from 1-day to 2-week. The t-statistics are shown in brackets.

2014-01-01 - 2023-12-31							
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7
Rt	-0.023	0.003	0.017	0.023**	0.021**	0.028***	0.021***
	(-1.02)	(0.19)	(1.39)	(2.25)	(2.31)	(3.57)	(2.74)
Adj. R^2	0.00	-0.00	0.00	0.00	0.00	0.00	0.00
	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.017**	0.016**	0.019***	0.019***	0.017***	0.015***	0.015***
	(2.41)	(2.35)	(3.06)	(3.10)	(2.90)	(2.76)	(2.89)
Adj. R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00
*** p<0.01, ** p<0.05, * p<0.1							

E.3. TSMOM by groups

Table .27: TSMOM by groups, Top 148 coins

Following Liu and Tsyvinski (2021), the two tables estimate the time-series momentum magnitude of the volume-weighted market return. Panel A presents the results by grouping volume-weighted market past cumulative returns into terciles labelled as Low, Middle, and High, and evaluating the subsequent day's return. Panel B groups the volume-weighted cumulative returns by the cutoffs calculated by the first two years' data (from 2014 to 2015), and evaluates the out-of-sample (from 2016 to 2023) TSMOM magnitude of the cryptocurrencies market. The results are presented in percentage expression and the t-statistics are reported in brackets. Diff refers to the difference in mean return between the High group and the Low group.

Panel A: Time-series momentum by groups (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-3.39*** (-34.04)	0.26** (2.00)	0.11 (0.81)	0.13 (0.98)	0.15 (1.16)	0.12 (0.93)	0.02 (0.17)	0.01 (0.06)	0.05 (0.36)
Middle	0.40*** (24.91)	0.43*** (3.78)	0.37*** (3.49)	0.35*** (3.45)	0.28*** (2.68)	0.32*** (2.94)	0.23** (2.22)	0.27*** (2.68)	0.34*** (3.40)
High	4.28*** (50.16)	0.54*** (4.47)	0.74*** (6.28)	0.73*** (6.06)	0.78*** (6.27)	0.78*** (6.42)	0.96*** (7.77)	0.94*** (7.55)	0.82*** (6.45)
Diff	7.67	0.28	0.63	0.60	0.63	0.66	0.94	0.94	0.78
Weekly HML	19.03*** (35.11)	0.68 (1.65)	1.56*** (3.59)	1.47*** (3.26)	1.51*** (3.21)	1.57*** (3.25)	2.25*** (4.79)	2.25*** (4.76)	1.87*** (3.96)
Panel B: Time-series momentum by groups with the first 2-year cutoffs (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-3.90*** (-31.10)	0.36** (2.14)	0.19 (1.03)	0.23 (1.17)	0.16 (0.88)	0.13 (0.67)	0.09 (0.44)	0.09 (0.41)	0.15 (0.73)
Middle	0.16*** (8.68)	0.56*** (4.64)	0.30*** (2.60)	0.35*** (3.14)	0.34*** (2.86)	0.44*** (3.65)	0.28** (2.40)	0.17* (1.66)	0.23** (2.01)
High	4.13*** (45.01)	0.54*** (4.21)	0.84*** (7.12)	0.78*** (6.53)	0.80*** (6.71)	0.73*** (6.39)	0.86*** (7.34)	0.92*** (7.89)	0.81*** (7.09)
Diff	8.02	0.18	0.65	0.54	0.63	0.60	0.77	0.83	0.67
Weekly HML	19.25*** (31.16)	0.73 (1.50)	2.03*** (4.04)	1.80*** (3.47)	2.11*** (3.82)	2.06*** (3.67)	2.55*** (4.69)	2.77*** (4.96)	2.50*** (4.49)

*** p<0.01, ** p<0.05, * p<0.1

E.4. WML portfolio performance

Table .28: WML long-short portfolios, Top 148 coins

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1%. The Sharpe ratio is annualised by 365 days.

Panel A: WML portfolio performance									
Lookback			1-day			7-day			14-day
	Winner	Loser	WML	Winner	Loser	WML	Winner	Loser	WML
Volume-weighted average return (daily, percentage)	0.66*** (6.79)	0.17* (1.89)	0.48*** (4.86)	0.68*** (7.60)	0.02 (-0.15)	0.66*** (7.27)	0.64*** (6.93)	0.16 (1.50)	0.48*** (5.39)
Sharpe ratio			1.30			1.88			1.32
Capital-weighted average return (daily, percentage)	0.24*** (4.29)	-0.04 (0.32)	0.29*** (3.90)	0.28*** (4.77)	-0.14* (-1.93)	0.41*** (6.03)	0.27*** (4.46)	0.00 (-0.02)	0.27*** (4.25)
Sharpe ratio			0.90			1.37			0.82
Equally-weighted average return (daily, percentage)	-0.07 (1.64)	0.07 (1.28)	-0.14 (0.59)	0.06*** (2.83)	-0.08 (-1.27)	0.14*** (3.89)	-0.01** (2.02)	-0.04 (-0.57)	0.03** (2.48)
Sharpe ratio			-0.51			0.50			0.09
Volatility-weighted average return (daily, percentage)	-	-	-	0.15*** (3.81)	-0.11* (-1.66)	0.26*** (5.29)	0.09*** (3.09)	-0.04 (-0.64)	0.13*** (3.66)
Sharpe ratio			-			1.02			0.48
Panel B: WML portfolio performance considering transaction costs									
Lookback			1-day			7-day			14-day
	Winner	Loser	WML	Winner	Loser	WML	Winner	Loser	WML
Volume-weighted average return (daily, percentage)	0.56*** (5.70)	0.07 (0.55)	0.28*** (2.84)	0.58*** (6.47)	-0.08 (-1.55)	0.46*** (5.13)	0.54*** (5.83)	0.06 (0.12)	0.28*** (3.31)
Sharpe ratio			0.76			1.30			0.77
Capital-weighted average return (daily, percentage)	0.14*** (3.19)	-0.14 (-1.08)	0.09* (1.77)	0.18*** (3.56)	-0.24*** (-3.38)	0.21*** (3.74)	0.17*** (3.31)	-0.10 (-1.48)	0.07** (2.06)
Sharpe ratio			0.26			0.70			0.19
Equally-weighted average return (daily, percentage)	-0.17 (0.52)	-0.03 (-0.06)	-0.34* (-1.69)	-0.04 (1.64)	-0.18*** (-2.65)	-0.06 (1.54)	-0.11 (0.86)	-0.14** (-1.97)	-0.17 (0.16)
Sharpe ratio			-1.24			-0.21			-0.59
Volatility-weighted average return (daily, percentage)	-	-	-	0.05** (2.57)	-0.21*** (-3.10)	0.06*** (2.79)	-0.01* (1.85)	-0.14** (-2.10)	-0.07 (1.16)
Sharpe ratio			-			0.23			-0.28

*** p<0.01, ** p<0.05, * p<0.1

E.5. WML portfolio after controlling for factors

Table .29: WML portfolios after controlling for factors, Top 148 coins

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

	1-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.17)	0.005*** (3.91)	0.006*** (5.51)	0.005*** (5.19)	0.003*** (2.76)	0.003** (2.57)	0.004*** (3.84)	0.003*** (3.59)	-0.001 (-1.52)	-0.001 (-1.38)	0.000 (-0.41)	0.000 (-0.45)
CMKT-Rf	0.02 (0.38)	0.02 (0.42)	0.01 (0.37)	0.02 (0.45)	0.01 (0.35)	0.02 (0.38)	0.01 (0.34)	0.01 (0.40)	-0.04 (-1.11)	-0.04 (-1.13)	-0.04 (-1.52)	-0.04 (-1.51)
CSMB		-0.06 (-0.85)		-0.11** (-2.22)		-0.05 (-0.97)		-0.09** (-2.43)		0.04 (0.65)		-0.01 (-0.32)
CMOM			0.67*** (8.67)	0.68*** (8.75)			0.55*** (6.64)	0.55*** (6.72)		0.75*** (10.85)	0.75*** (10.87)	
Adj. R^2	0.00	0.00	0.20	0.21	0.00	0.00	0.18	0.19	0.00	0.00	0.45	0.45
	7-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.006*** (5.84)	0.006*** (5.59)	0.006*** (6.55)	0.006*** (6.31)	0.004*** (4.26)	0.004*** (4.00)	0.004*** (4.58)	0.004*** (4.32)	0.002* (1.70)	0.002* (1.68)	0.001** (2.00)	0.001* (1.95)
CMKT-Rf	0.06 (1.24)	0.06 (1.29)	0.08** (1.98)	0.09** (2.06)	0.04 (1.00)	0.05 (1.07)	0.06 (1.55)	0.07* (1.65)	-0.05 (-1.20)	-0.05 (-1.20)	-0.02 (-0.50)	-0.02 (-0.49)
CSMB		-0.08 (-1.13)		-0.10** (-2.14)		-0.11* (-1.74)		-0.12*** (-2.67)		0.00 (0.06)		-0.02 (-0.43)
CMOM			0.58*** (12.56)	0.59*** (12.58)			0.44*** (9.50)	0.44*** (9.69)		0.70*** (18.97)	0.70*** (18.87)	
Adj. R^2	0.00	0.00	0.17	0.17	0.00	0.01	0.13	0.14	0.00	0.00	0.37	0.37
	14-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (3.83)	0.005*** (3.68)	0.005*** (4.59)	0.005*** (4.45)	0.003*** (2.49)	0.003** (2.34)	0.003*** (3.06)	0.003*** (2.90)	0.000 (0.39)	0.001 (0.52)	0.001 (1.02)	0.001 (1.13)
CMKT-Rf	0.04 (0.92)	0.04 (0.92)	0.06 (1.37)	0.06 (1.38)	0.05 (1.13)	0.05 (1.13)	0.06 (1.52)	0.06 (1.53)	-0.06 (-1.49)	-0.07 (-1.49)	-0.05 (-1.28)	-0.05 (-1.29)
CSMB		0.00 (-0.02)		-0.02 (-0.35)		-0.01 (-0.11)		-0.03 (-0.38)		0.07 (0.79)		0.05 (0.82)
CMOM			0.64*** (12.63)	0.64*** (12.71)			0.50*** (10.79)	0.50*** (10.90)		0.68*** (15.09)	0.68*** (15.09)	
Adj. R^2	0.00	0.00	0.18	0.18	0.00	0.00	0.14	0.14	0.00	0.00	0.32	0.32

*** p<0.01, ** p<0.05, * p<0.1

Table .30: WML portfolios with transaction cost after controlling for factors, Top 148 coins

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

	1-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.003** (2.42)	0.003** (2.23)	0.004*** (3.59)	0.004*** (3.31)	0.001 (0.78)	0.001 (0.66)	0.002* (1.69)	0.001 (1.47)	-0.003*** (-3.74)	-0.003*** (-3.54)	-0.002*** (-3.37)	-0.002*** (-3.39)
CMKT-Rf	0.02 (0.38)	0.02 (0.42)	0.01 (0.37)	0.02 (0.45)	0.01 (0.35)	0.02 (0.38)	0.01 (0.34)	0.01 (0.40)	-0.04 (-1.11)	-0.04 (-1.13)	-0.04 (-1.52)	-0.04 (-1.51)
CSMB	-0.06 (-0.85)	-0.06 (-0.85)	-0.11** (-2.22)	-0.11** (-2.22)	-0.05 (-0.97)	-0.05 (-0.97)	-0.09** (-2.43)	-0.09** (-2.43)	0.04 (0.65)	0.04 (0.65)	0.04 (0.65)	-0.01 (-0.32)
CMOM	0.67*** (8.67)	0.67*** (8.67)	0.68*** (8.75)	0.68*** (8.75)	0.55*** (6.64)	0.55*** (6.64)	0.55*** (6.72)	0.55*** (6.72)	0.75*** (10.85)	0.75*** (10.85)	0.75*** (10.85)	0.75*** (10.87)
Adj. R^2	0.00	0.00	0.20	0.21	0.00	0.00	0.18	0.19	0.00	0.00	0.45	0.45
	7-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.004*** (4.02)	0.004*** (3.80)	0.004*** (4.44)	0.004*** (4.21)	0.002** (2.15)	0.002* (1.90)	0.002*** (2.22)	0.002* (1.93)	-0.001 (-0.62)	-0.001 (-0.60)	-0.001 (-1.30)	-0.001 (-1.36)
CMKT-Rf	0.06 (1.24)	0.06 (1.29)	0.08** (1.98)	0.09** (2.06)	0.04 (1.00)	0.05 (1.07)	0.06 (1.55)	0.07* (1.65)	-0.05 (-1.20)	-0.05 (-1.20)	-0.02 (-0.50)	-0.02 (-0.49)
CSMB	-0.08 (-1.13)	-0.08 (-1.13)	-0.10** (-2.14)	-0.10** (-2.14)	-0.11* (-1.74)	-0.11* (-1.74)	-0.12*** (-2.67)	-0.12*** (-2.67)	0.00 (0.06)	0.00 (0.06)	0.00 (0.06)	-0.02 (-0.43)
CMOM	0.58*** (12.56)	0.58*** (12.56)	0.59*** (12.58)	0.59*** (12.58)	0.44*** (9.50)	0.44*** (9.50)	0.44*** (9.69)	0.44*** (9.69)	0.70*** (18.97)	0.70*** (18.97)	0.70*** (18.97)	0.70*** (18.87)
Adj. R^2	0.00	0.00	0.17	0.17	0.00	0.01	0.13	0.14	0.00	0.00	0.37	0.37
	14-day formation period											
	VW-WML				CW-WML				EQ-WML			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.003** (2.20)	0.003** (2.11)	0.003*** (2.78)	0.003*** (2.68)	0.001 (0.54)	0.001 (0.49)	0.001 (0.91)	0.001 (0.83)	-0.002* (-1.76)	-0.002 (-1.52)	-0.001 (-1.63)	-0.001 (-1.47)
CMKT-Rf	0.04 (0.92)	0.04 (0.92)	0.06 (1.37)	0.06 (1.38)	0.05 (1.13)	0.05 (1.13)	0.06 (1.52)	0.06 (1.53)	-0.06 (-1.49)	-0.07 (-1.49)	-0.05 (-1.28)	-0.05 (-1.29)
CSMB	0.00 (-0.02)	0.00 (-0.02)	-0.02 (-0.35)	-0.02 (-0.35)	-0.01 (-0.11)	-0.01 (-0.11)	-0.03 (-0.38)	-0.03 (-0.38)	0.07 (0.79)	0.07 (0.79)	0.05 (0.82)	0.05 (0.82)
CMOM	0.64*** (12.63)	0.64*** (12.63)	0.64*** (12.71)	0.64*** (12.71)	0.50*** (10.79)	0.50*** (10.79)	0.50*** (10.90)	0.50*** (10.90)	0.68*** (15.09)	0.68*** (15.09)	0.68*** (15.09)	0.68*** (15.84)
Adj. R^2	0.00	0.00	0.18	0.18	0.00	0.00	0.14	0.14	0.00	0.00	0.32	0.32
Adj. R^2	0.00	0.00	0.21	0.21	0.00	0.00	0.21	0.21	0.00	0.00	0.21	0.21

*** p<0.01, ** p<0.05, * p<0.1

E.6. Winner portfolios after controlling for factors

Table .31: Winner portfolios after controlling for factors, Top 148 coins

The table shows the regression results of the Winner portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted Winner portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

	1-day formation period											
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (5.51)	0.005*** (5.94)	0.006*** (6.51)	0.006*** (6.88)	0.001 (1.31)	0.001 (1.60)	0.002*** (2.01)	0.002*** (2.24)	-0.002*** (-2.30)	-0.001 (-1.55)	-0.001* (-1.69)	-0.001 (-0.81)
CMKT-Rf	1.01*** (32.26)	1.00*** (31.59)	1.00*** (34.87)	1.00*** (34.49)	0.95*** (26.69)	0.94*** (26.14)	0.94*** (28.46)	0.94*** (28.04)	0.94*** (26.57)	0.93*** (28.66)	0.94*** (28.35)	0.93*** (31.51)
CSMB	0.15*** (3.65)	0.15*** (3.65)	0.15*** (3.83)	0.15*** (3.83)	0.11*** (3.34)	0.11*** (3.34)	0.11*** (3.30)	0.11*** (3.30)	0.42*** (9.62)	0.42*** (9.62)	0.38*** (9.37)	0.38*** (9.37)
CMOM	0.46*** (5.42)	0.46*** (5.42)	0.46*** (5.34)	0.45*** (5.34)	0.38*** (4.20)	0.38*** (4.20)	0.37*** (4.14)	0.37*** (4.14)	0.50*** (6.12)	0.50*** (6.12)	0.47*** (5.81)	0.47*** (5.81)
Adj. R^2	0.33	0.34	0.43	0.44	0.36	0.36	0.44	0.45	0.37	0.48	0.53	0.62

	7-day formation period											
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (6.16)	0.006*** (6.60)	0.005*** (6.61)	0.005*** (7.07)	0.001* (1.90)	0.002*** (2.15)	0.001* (1.92)	0.002*** (2.16)	-0.001 (-0.81)	0.000 (0.24)	-0.001 (-1.18)	0.000 (-0.02)
CMKT-Rf	0.99*** (27.39)	0.99*** (27.53)	1.01*** (28.25)	1.01*** (28.67)	0.93*** (26.82)	0.93*** (26.86)	0.95*** (27.59)	0.94*** (27.77)	0.91*** (26.12)	0.90*** (27.89)	0.93*** (27.49)	0.92*** (30.81)
CSMB	0.13*** (3.63)	0.13*** (3.63)	0.12*** (4.57)	0.12*** (4.57)	0.08*** (2.79)	0.08*** (2.79)	0.07*** (2.88)	0.07*** (2.88)	0.39*** (10.59)	0.39*** (10.59)	0.38*** (14.08)	0.38*** (14.08)
CMOM	0.35*** (9.61)	0.35*** (9.61)	0.34*** (10.05)	0.34*** (10.05)	0.25*** (6.54)	0.25*** (6.54)	0.24*** (6.60)	0.24*** (6.60)	0.41*** (13.48)	0.41*** (13.48)	0.40*** (15.53)	0.40*** (15.53)
Adj. R^2	0.34	0.35	0.40	0.41	0.38	0.38	0.42	0.42	0.35	0.44	0.46	0.54

	14-day formation period											
	VW-Winner				CW-Winner				EQ-Winner			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.005*** (4.77)	0.005*** (5.26)	0.005*** (5.47)	0.006*** (6.04)	0.001 (1.60)	0.002*** (2.01)	0.002*** (1.99)	0.002*** (2.42)	-0.001 (-1.53)	-0.001 (-0.61)	-0.001 (-1.40)	0.000 (-0.37)
CMKT-Rf	0.99*** (25.99)	0.99*** (25.53)	1.00*** (27.54)	1.00*** (27.33)	0.95*** (25.47)	0.95*** (25.14)	0.96*** (26.81)	0.95*** (26.65)	0.91*** (22.74)	0.90*** (23.54)	0.92*** (24.58)	0.91*** (26.13)
CSMB	0.20*** (3.00)	0.20*** (3.00)	0.19*** (3.81)	0.19*** (3.81)	0.17*** (2.42)	0.17*** (2.42)	0.16*** (2.83)	0.16*** (2.83)	0.44*** (7.44)	0.44*** (7.44)	0.42*** (10.07)	0.42*** (10.07)
CMOM	0.44*** (8.63)	0.44*** (8.63)	0.44*** (9.64)	0.44*** (9.64)	0.34*** (7.03)	0.34*** (7.03)	0.33*** (7.82)	0.33*** (7.82)	0.44*** (9.54)	0.44*** (9.54)	0.42*** (12.49)	0.42*** (12.49)
Adj. R^2	0.32	0.34	0.41	0.43	0.34	0.36	0.41	0.42	0.33	0.43	0.44	0.53

*** p<0.01, ** p<0.05, * p<0.1

F. Exclude Bitcoin

F.1. Summary statistics

Table .32: Summary statistics for research sample excluding Bitcoin

The two tables tabulate the summary statistics of our research data excluding Bitcoin. Panel A provides the number of coins and the mean and median of market capitalisation and daily trading volume in USD by year. We only consider coins with a market capitalisation above \$1 million daily sampling from January 2014 to December 2023. Panel B shows the summary statistics of the daily return of the volume-weighted market return, Bitcoin, Ethereum, and XRP.

Panel A: Sample data					
Year	Count	Market Cap (mil)		Volume (thous)	
		Mean	Median	Mean	Median
2014	52	31.09	4.97	454.07	35.42
2015	46	24.82	4.60	227.35	16.50
2016	96	37.59	4.28	754.05	26.40
2017	396	259.64	13.50	12,079.73	209.29
2018	827	345.94	15.38	17,331.42	249.44
2019	972	120.12	6.26	30,862.07	224.29
2020	1325	153.19	7.42	52,311.80	343.20
2021	2324	629.59	14.89	79,042.61	701.94
2022	2141	441.28	11.23	34,961.20	492.45
2023	2024	394.04	12.95	18,715.91	425.17
Full	3191	243.65	8.70	24,675.04	260.60
Panel B: Volume-weighted market return and major cryptocurrencies					
	Mean	Median	STD	Skewness	Kurtosis
VW market	1.03%	0.75%	0.06	-0.23	14.05
CW market	0.35%	0.31%	0.05	1.85	38.54
EQ market	0.34%	0.51%	0.04	-1.18	10.79
Bitcoin	0.11%	0.11%	0.04	-0.76	10.95
Ethereum	0.26%	0.04%	0.06	0.01	8.74
XRP	0.09%	-0.17%	0.06	1.81	35.18

F.2. TSMOM regression

Table .33: TSMOM of the volume-weighted market return, Exclude Bitcoin

This table presents the TSMOM results for the volume-weighted market return by regressing the current market return on the past cumulative market return among different time horizons. We show the results of cumulative market return from 1-day to 2-week. The t-statistics are shown in brackets.

2014-01-01 - 2023-12-31							
	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7
Rt	0.050 (1.60)	0.046** (2.34)	0.055*** (3.74)	0.051*** (4.45)	0.044*** (4.52)	0.046*** (5.47)	0.039*** (5.12)
Adj. R^2	0.00	0.00	0.01	0.01	0.01	0.02	0.01
	Rt-1:t-8	Rt-1:t-9	Rt-1:t-10	Rt-1:t-11	Rt-1:t-12	Rt-1:t-13	Rt-1:t-14
Rt	0.035*** (5.11)	0.033*** (5.16)	0.033*** (5.52)	0.032*** (5.59)	0.029*** (5.42)	0.028*** (5.51)	0.028*** (5.73)
Adj. R^2	0.01	0.01	0.02	0.02	0.02	0.02	0.02
*** p<0.01, ** p<0.05, * p<0.1							

F.3. TSMOM by groups

Table .34: TSMOM by groups, Exclude Bitcoin

Following Liu and Tsyvinski (2021), the two tables estimate the time-series momentum magnitude of the volume-weighted market return. Panel A presents the results by grouping volume-weighted market past cumulative returns into terciles labelled as Low, Middle, and High, and evaluating the subsequent day's return. Panel B groups the volume-weighted cumulative returns by the cutoffs calculated by the first two years' data (from 2014 to 2015), and evaluates the out-of-sample (from 2016 to 2023) TSMOM magnitude of the cryptocurrencies market. The results are presented in percentage expression and the t-statistics are reported in brackets. Diff refers to the difference in mean return between the High group and the Low group.

Panel A: Time-series momentum by groups (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-4.63***	0.64***	0.39*	0.28	0.24	0.39**	0.30	0.25	0.24
	(-31.61)	(3.35)	(1.92)	(1.42)	(1.27)	(2.08)	(1.62)	(1.33)	(1.35)
Middle	0.81***	0.81***	0.83***	0.78***	0.84***	0.60***	0.50***	0.69***	0.67***
	(30.85)	(5.39)	(6.18)	(5.64)	(5.80)	(4.16)	(3.56)	(4.96)	(4.71)
High	6.92***	1.65***	1.85***	2.01***	1.99***	2.09***	2.27***	2.15***	2.17***
	(47.89)	(8.35)	(9.59)	(10.48)	(10.19)	(10.31)	(11.05)	(10.55)	(10.50)
Diff	11.55	1.01	1.46	1.73	1.75	1.70	1.97	1.90	1.93
Weekly HML	26.94***	2.35***	3.41***	4.03***	4.07***	3.98***	4.60***	4.44***	4.50***
	(32.08)	(3.42)	(4.72)	(5.27)	(5.24)	(4.92)	(5.79)	(5.57)	(5.64)
Panel B: Time-series momentum by groups with the first 2-year cutoffs (percentage)									
Rank	Rt	Rt-1	Rt-1:t-2	Rt-1:t-3	Rt-1:t-4	Rt-1:t-5	Rt-1:t-6	Rt-1:t-7	Rt-1:t-14
Low	-5.54***	0.73***	0.45	0.31	0.23	0.33	0.16	0.27	0.15
	(-27.63)	(2.79)	(1.47)	(1.04)	(0.83)	(1.07)	(0.55)	(0.86)	(0.57)
Middle	0.21***	0.85***	0.72***	0.64***	0.69***	0.59***	0.53***	0.37**	0.62***
	(7.14)	(5.45)	(4.77)	(3.83)	(4.38)	(3.73)	(3.72)	(2.48)	(4.25)
High	6.00***	1.67***	1.92***	1.97***	1.93***	1.93***	1.99***	2.07***	2.02***
	(43.57)	(8.83)	(11.01)	(11.89)	(11.17)	(11.26)	(11.41)	(12.11)	(10.99)
Diff	11.54	0.94	1.47	1.66	1.70	1.60	1.83	1.80	1.87
Weekly HML	27.20***	3.54***	4.87***	5.63***	5.87***	5.79***	6.47***	6.49***	6.27***
	(28.99)	(4.28)	(5.87)	(6.46)	(6.48)	(6.25)	(7.05)	(7.08)	(6.71)

*** p<0.01, ** p<0.05, * p<0.1

F.4. WML portfolio performance

Table .35: WML long-short portfolios, Exclude Bitcoin

This table presents the mean daily return of the volume-weighted WML portfolio compared to the capital-weighted, equally-weighted, and inverse volatility-weighted WML portfolios. Panel A reports the average return of WML portfolios and Panel B provides the WML returns considering fixed transaction costs of 0.1%. The Sharpe ratio is annualised by 365 days.

Panel A: WML portfolio performance									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.49*** (10.90)	0.34*** (2.86)	1.15*** (7.76)	1.39*** (11.40)	0.30*** (3.09)	1.09*** (9.14)	1.33*** (10.70)	0.41*** (4.04)	0.92*** (7.34)
Sharpe ratio			2.44			2.88			2.31
Capital-weighted average return (daily, percentage)	0.52*** (4.23)	0.21** (2.22)	0.31** (2.54)	0.47*** (4.44)	0.09 (1.01)	0.38*** (3.57)	0.50*** (4.62)	0.11 (1.32)	0.39*** (3.92)
Sharpe ratio			0.79			1.11			1.22
Equally-weighted average return (daily, percentage)	-0.77*** (-9.88)	0.43*** (6.01)	-1.20*** (-21.86)	-0.71*** (-9.17)	0.20*** (2.93)	-0.92*** (-16.59)	-0.63*** (-8.09)	0.18** (2.57)	-0.81*** (-14.47)
Sharpe ratio			-6.93			-5.27			-4.60
Volatility-weighted average return (daily, percentage)	-	-	-	-0.13** (-2.01)	-0.05 (-0.84)	-0.08 (-1.35)	-0.18** (-2.49)	-0.01 (-0.17)	-0.16*** (-2.67)
Sharpe ratio			-			-0.45			-0.87
Panel B: WML portfolio performance considering transaction cost of 10 bps									
Lookback	Winner	Loser	1-day WML	Winner	Loser	7-day WML	Winner	Loser	14-day WML
Volume-weighted average return (daily, percentage)	1.39*** (10.16)	0.24** (2.01)	0.95*** (6.41)	1.29*** (10.58)	0.20** (2.05)	0.89*** (7.46)	1.23*** (9.89)	0.31*** (3.06)	0.72*** (5.74)
Sharpe ratio			2.02			2.35			1.80
Capital-weighted average return (daily, percentage)	0.42*** (3.42)	0.11 (1.14)	0.11 (0.91)	0.37*** (3.50)	-0.01 (-0.10)	0.18* (1.70)	0.40*** (3.69)	0.01 (0.08)	0.19* (1.91)
Sharpe ratio			0.27			0.52			0.59
Equally-weighted average return (daily, percentage)	-0.87*** (-11.16)	0.33*** (4.61)	-1.40*** (-25.50)	-0.81*** (-10.46)	0.1 (1.49)	-1.12*** (-20.21)	-0.73*** (-9.38)	0.08 (1.11)	-1.01*** (-18.06)
Sharpe ratio			-8.09			-6.42			-5.73
Volatility-weighted average return (daily, percentage)	-	-	-	-0.23*** (-3.53)	-0.15** (-2.40)	-0.28*** (-4.76)	-0.28*** (-3.92)	-0.11* (-1.74)	-0.36*** (-5.92)
Sharpe ratio			-			-1.53			-1.90

*** p<0.01, ** p<0.05, * p<0.1

F.5. WML portfolio after controlling for factors

Table .36: WML portfolios after controlling for factors - Exclude Bitcoin

The table shows the regression results of the WML portfolio excess returns after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.011*** (7.85)	0.011*** (7.91)	0.026*** (12.38)	0.027*** (12.48)	0.003** (2.54)	0.003*** (2.62)	0.012*** (9.03)	0.013*** (8.99)	-0.012*** (-18.88)	-0.012*** (-19.02)	0.000 (-0.85)	0.000 (-0.75)	-	-	-	-
CMKT-Rf	0.08 (1.43)	0.04 (0.70)	0.09* (1.66)	0.03 (0.49)	0.15 (1.44)	0.10 (1.32)	0.15 (1.48)	0.09 (1.22)	0.01 (0.76)	0.02 (1.12)	0.02 (1.43)	0.01 (0.77)				
CSMB	-0.09* (-1.81)			-0.13*** (-2.94)	-0.09 (-1.27)			-0.12* (-1.65)		0.02 (0.78)		-0.01 (-1.08)				
CMOM			0.95*** (11.71)	0.96*** (11.90)			0.60*** (11.16)	0.61*** (11.52)			0.72*** (33.85)	0.72*** (34.19)				
Adj. R^2	0.00	0.00	0.14	0.15	0.01	0.01	0.09	0.10	0.00	0.00	0.59	0.59				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.010*** (7.96)	0.011*** (8.06)	0.021*** (12.45)	0.021*** (12.47)	0.004*** (3.52)	0.004*** (3.49)	0.011*** (9.37)	0.011*** (9.10)	-0.009*** (-14.74)	-0.009*** (-14.53)	0.000 (0.07)	0.000 (0.05)	0.000 (-0.63)	0.000 (-0.41)	0.005*** (7.47)	0.005*** (7.43)
CMKT-Rf	0.12*** (2.73)	0.08* (1.75)	0.15*** (3.83)	0.10*** (2.69)	0.07 (0.88)	0.04 (0.62)	0.09 (1.10)	0.05 (0.89)	-0.01 (-0.45)	0 (-0.16)	0.01 (1.22)	0.01 (1.24)	-0.13*** (-4.27)	-0.15*** (-4.04)	-0.11*** (-4.65)	-0.14*** (-4.60)
CSMB	-0.10* (-1.89)			-0.11** (-2.29)	-0.07 (-1.01)			-0.08 (-1.14)		0.01 (0.41)		0.00 (0.18)		-0.04 (-1.43)	-0.05* (-1.81)	-0.05* (-1.81)
CMOM			0.83*** (12.20)	0.83*** (12.21)			0.59*** (11.34)	0.60*** (11.38)			0.72*** (34.64)	0.72*** (34.49)			0.45*** (16.19)	0.45*** (16.27)
Adj. R^2	0.01	0.01	0.15	0.16	0.00	0.01	0.10	0.10	0.00	0.00	0.52	0.52	0.04	0.04	0.22	0.22
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.009*** (6.54)	0.009*** (6.85)	0.018*** (10.19)	0.018*** (10.43)	0.003*** (3.35)	0.004*** (4.01)	0.010*** (8.26)	0.010*** (8.82)	-0.008*** (-12.42)	-0.008*** (-12.19)	0.000 (0.07)	0.000 (0.15)	-0.001** (-2.01)	-0.001** (-1.65)	0.004*** (5.78)	0.004*** (5.99)
CMKT-Rf	0.15*** (3.26)	0.08* (1.77)	0.16*** (3.97)	0.09** (2.48)	0.22*** (3.15)	0.12** (2.44)	0.23*** (3.37)	0.13*** (2.82)	0.01 (0.65)	0 (-0.15)	0.02** (2.20)	0.01 (1.13)	-0.10*** (-3.36)	-0.13*** (-3.86)	-0.09*** (-3.48)	-0.12*** (-3.93)
CSMB	-0.16*** (-3.24)			-0.15*** (-3.37)	-0.22*** (-3.42)			-0.21*** (-3.42)		-0.03 (-1.46)		-0.02 (-1.43)		-0.07*** (-2.88)	-0.07*** (-3.03)	-0.07*** (-3.03)
CMOM			0.84*** (10.20)	0.84*** (10.27)			0.61*** (10.94)	0.60*** (11.15)			0.75*** (26.06)	0.75*** (26.14)			0.52*** (17.98)	0.52*** (18.25)
Adj. R^2	0.01	0.02	0.15	0.15	0.04	0.06	0.15	0.17	0.00	0.00	0.54	0.54	0.02	0.03	0.23	0.24
*** p<0.01, ** p<0.05, * p<0.1																

*** p<0.01, ** p<0.05, * p<0.1

Table .37: WML portfolios with transaction cost after controlling for factors - Exclude Bitcoin

The table shows the regression outcomes of the WML portfolio excess returns minus transaction cost of 0.1% after controlling for standard cryptocurrency risk factors, including CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted WML portfolio needs to weigh by the volatility of more than one day, hence, there is no data for the 1-day formation period.

1-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.009*** (6.44)	0.009*** (6.52)	0.025*** (11.44)	0.025*** (11.55)	0.001 (0.56)	0.001 (0.77)	0.010*** (7.56)	0.011*** (7.58)	-0.014*** (-22.01)	-0.014*** (-22.16)	-0.002*** (-4.68)	-0.002*** (-4.58)	-	-	-	-
CMKT-Rf	0.08 (1.43)	0.04 (0.70)	0.09* (1.66)	0.03 (0.49)	0.15 (1.44)	0.10 (1.32)	0.15 (1.48)	0.09 (1.22)	0.01 (0.76)	0.02 (1.12)	0.02 (1.43)	0.01 (0.77)				
CSMB		-0.09* (-1.81)		-0.13*** (-2.94)	-0.09 (-1.27)		-0.12* (-1.65)			0.02 (0.78)		-0.01 (-1.08)				
CMOM			0.95*** (11.71)	0.96*** (11.90)			0.60*** (11.16)	0.61*** (11.52)			0.72*** (33.85)	0.72*** (34.19)				
Adj. R^2	0.00	0.00	0.14	0.15	0.01	0.01	0.09	0.10	0.00	0.00	0.59	0.59				
7-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.008*** (6.43)	0.009*** (6.56)	0.019*** (11.26)	0.019*** (11.30)	0.002 (1.52)	0.002 (1.63)	0.009*** (7.68)	0.009*** (7.49)	-0.011*** (-17.95)	-0.011*** (-17.69)	-0.002*** (-4.05)	-0.002*** (-3.94)	-0.002*** (-3.76)	-0.002*** (-3.43)	0.003*** (4.69)	0.004*** (4.75)
CMKT-Rf	0.12*** (2.73)	0.08* (1.75)	0.15*** (3.83)	0.10*** (2.69)	0.07 (0.88)	0.04 (0.62)	0.09 (1.10)	0.05 (0.89)	-0.01 (-0.45)	0.00 (-0.16)	0.01 (1.22)	0.01 (1.24)	-0.13*** (-4.27)	-0.15*** (-4.04)	-0.11*** (-4.65)	-0.14*** (-4.60)
CSMB		-0.10* (-1.89)		-0.11** (-2.29)	-0.07 (-1.01)		-0.08 (-1.14)			0.01 (0.41)		0.00 (0.18)		-0.04 (-1.43)		-0.05* (-1.81)
CMOM			0.83*** (12.20)	0.83*** (12.21)			0.59*** (11.34)	0.60*** (11.38)		0.72*** (34.64)	0.72*** (34.49)	0.45*** (16.19)		0.45*** (16.27)		0.45*** (16.27)
Adj. R^2	0.01	0.01	0.15	0.16	0.00	0.01	0.10	0.10	0.00	0.00	0.52	0.52	0.04	0.04	0.22	0.22
14-day formation period																
VW-WML				CW-WML				EQ-WML				VOL-WML				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.007*** (5.02)	0.007*** (5.34)	0.016*** (9.05)	0.016*** (9.29)	0.001 (1.18)	0.002* (1.85)	0.008*** (6.56)	0.008*** (7.11)	-0.010*** (-15.47)	-0.010*** (-15.22)	-0.002*** (-3.29)	-0.002*** (-3.23)	-0.003*** (-5.00)	-0.003*** (-4.57)	0.002*** (3.11)	0.003*** (3.33)
CMKT-Rf	0.15*** (3.26)	0.08* (1.77)	0.16*** (3.97)	0.09** (2.48)	0.22*** (3.15)	0.12*** (2.44)	0.23*** (3.37)	0.13*** (2.82)	0.01 (0.65)	0 (-0.15)	0.02** (2.20)	0.01 (1.13)	-0.10*** (-3.36)	-0.13*** (-3.86)	-0.09*** (-3.48)	-0.12*** (-3.93)
CSMB		-0.16*** (-3.24)		-0.15*** (-3.37)	-0.22*** (-3.42)		-0.21*** (-3.42)			-0.03 (-1.46)		-0.02 (-1.43)		-0.07*** (-2.88)		-0.07*** (-3.03)
CMOM			0.84*** (10.20)	0.84*** (10.27)			0.61*** (10.94)	0.60*** (11.15)		0.75*** (26.06)	0.75*** (26.14)	0.52*** (17.98)		0.52*** (17.98)		0.52*** (18.25)
Adj. R^2	0.01	0.02	0.15	0.15	0.04	0.06	0.15	0.17	0.00	0.00	0.54	0.54	0.02	0.03	0.23	0.24
*** p<0.01, ** p<0.05, * p<0.1																

*** p<0.01, ** p<0.05, * p<0.1

F.6. Winner portfolios after controlling for factors

Table .38: Winner portfolios after controlling for factors - Exclude Bitcoin

The table shows the regression results of the Winner portfolio excess returns after controlling for standard cryptocurrency risk factors, CMKT, CSMB, and CMOM factors. The t-statistics are reported in parentheses. The inverse volatility-weighted Winner portfolio needs to weigh by the volatility of more than one day, thus, there is no data for the 1-day formation period.

1-day formation period																
VW Winner				CW Winner				EQ Winner				VOL Winner				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.012*** (9.94)	0.011*** (9.59)	0.022*** (11.41)	0.021*** (11.10)	0.002** (2.48)	0.007*** (6.35)	0.007*** (5.86)	0.007*** (5.86)	-0.010*** (-13.91)	-0.011*** (-19.02)	-0.003*** (-3.66)	-0.004*** (-6.86)	-	-	-	-
CMKT-Rf	0.86*** (9.73)	0.96*** (16.58)	0.86*** (9.87)	0.95*** (16.52)	0.91*** (9.12)	0.94*** (13.14)	0.91*** (9.15)	0.94*** (13.03)	0.59*** (10.23)	0.78*** (21.27)	0.59*** (10.19)	0.77*** (20.80)	0.77*** (20.80)	0.77*** (20.80)	0.77*** (20.80)	0.77*** (20.80)
CSMB	0.21*** (3.02)	0.21*** (3.02)	0.19*** (2.68)	0.19*** (2.68)	0.08 (1.08)	0.08 (1.08)	0.06 (0.88)	0.06 (0.88)	0.42*** (7.11)	0.42*** (7.11)	0.42*** (7.11)	0.40*** (6.85)	0.40*** (6.85)	0.40*** (6.85)	0.40*** (6.85)	0.40*** (6.85)
CMOM			0.62*** (8.50)	0.61*** (8.35)		0.35*** (7.32)	0.34*** (7.18)	0.34*** (7.18)			0.43*** (16.81)	0.40*** (17.74)	0.43*** (16.81)	0.40*** (17.74)	0.43*** (16.81)	0.40*** (17.74)
Adj. R^2	0.31	0.32	0.38	0.39	0.43	0.43	0.45	0.46	0.43	0.59	0.54	0.68	0.43	0.59	0.54	0.68
7-day formation period																
VW Winner				CW Winner				EQ Winner				VOL Winner				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.011*** (10.11)	0.010*** (9.43)	0.017*** (11.52)	0.017*** (11.05)	0.002** (2.54)	0.002** (1.95)	0.006*** (6.44)	0.006*** (5.71)	-0.009*** (-12.38)	-0.010*** (-16.00)	-0.004*** (-4.70)	-0.005*** (-7.46)	-0.003*** (-3.96)	-0.004*** (-5.46)	0.001 (0.73)	0.000 (-0.47)
CMKT-Rf	0.84*** (9.62)	0.93*** (17.20)	0.86*** (9.78)	0.95*** (17.57)	0.80*** (9.33)	0.85*** (15.60)	0.82*** (9.32)	0.86*** (15.58)	0.56*** (9.66)	0.74*** (21.69)	0.57*** (9.73)	0.75*** (21.79)	0.37*** (8.09)	0.50*** (13.49)	0.38*** (8.32)	0.51*** (14.38)
CSMB	0.21*** (3.15)	0.21*** (3.15)	0.20*** (3.08)	0.20*** (3.08)	0.11 (1.58)	0.11 (1.58)	0.1 (1.56)	0.1 (1.56)	0.40*** (8.06)	0.40*** (8.06)	0.39*** (7.81)	0.39*** (7.81)	0.29*** (7.78)	0.29*** (7.78)	0.29*** (7.71)	0.29*** (7.71)
CMOM			0.50*** (7.99)	0.49*** (8.04)		0.33*** (6.92)	0.33*** (6.91)	0.33*** (6.91)			0.43*** (14.99)	0.42*** (16.52)	0.25*** (8.75)	0.25*** (8.75)	0.25*** (8.75)	0.25*** (8.74)
Adj. R^2	0.36	0.38	0.42	0.43	0.44	0.44	0.47	0.47	0.39	0.53	0.49	0.63	0.24	0.35	0.29	0.39
14-day formation period																
VW Winner				CW Winner				EQ Winner				VOL Winner				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Intercept	0.010*** (9.62)	0.010*** (9.12)	0.016*** (10.09)	0.015*** (9.74)	0.002*** (2.72)	0.002*** (2.56)	0.005*** (5.42)	0.005*** (5.27)	-0.008*** (-11.23)	-0.009*** (-14.09)	-0.004*** (-4.57)	-0.004*** (-6.38)	-0.003*** (-4.61)	-0.004*** (-6.03)	0.000 (-0.20)	-0.001 (-1.15)
CMKT-Rf	0.85*** (10.05)	0.92*** (16.50)	0.86*** (10.09)	0.93*** (16.77)	0.87*** (11.69)	0.88*** (17.79)	0.88*** (11.71)	0.89*** (17.94)	0.57*** (10.17)	0.73*** (20.74)	0.58*** (10.10)	0.74*** (20.81)	0.40*** (8.61)	0.52*** (13.63)	0.41*** (8.64)	0.53*** (13.97)
CSMB	0.14*** (2.16)	0.14*** (2.16)	0.15*** (2.31)	0.15*** (2.31)	0.03 (0.40)	0.03 (0.40)	0.03 (0.52)	0.03 (0.52)	0.35*** (6.56)	0.35*** (6.56)	0.36*** (6.71)	0.36*** (6.71)	0.27*** (6.14)	0.27*** (6.14)	0.27*** (6.31)	0.27*** (6.31)
CMOM			0.51*** (6.36)	0.52*** (6.37)		0.33*** (6.37)	0.33*** (6.38)	0.33*** (6.38)			0.44*** (13.37)	0.44*** (13.82)	0.28*** (9.58)	0.28*** (9.58)	0.28*** (9.58)	0.29*** (9.78)
Adj. R^2	0.36	0.37	0.42	0.42	0.50	0.50	0.54	0.54	0.41	0.52	0.51	0.62	0.25	0.33	0.30	0.38
*** p<0.01, ** p<0.05, * p<0.1																

*** p<0.01, ** p<0.05, * p<0.1